

ADULT DRUG COURT BEST PRACTICE STANDARDS

VOLUME II

**NATIONAL ASSOCIATION OF DRUG COURT PROFESSIONALS
ALEXANDRIA, VIRGINIA**

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THE NATIONAL ASSOCIATION OF DRUG COURT PROFESSIONALS

It takes innovation, passion, teamwork, and strong judicial leadership for a community to achieve success in rehabilitating persons with severe substance use disorders and concurrent criminal involvement. That is why since 1994, the National Association of Drug Court Professionals (NADCP) has worked tirelessly at the national, state, and local levels to develop and enhance Drug Courts, which combine treatment and accountability to support and compel drug-addicted persons charged with serious crimes to change their lives.

Now an international movement, Drug Courts are the shining example of what works in the criminal justice system. Today over 2,900 Drug Courts operate in the U.S. and another thirteen countries have also implemented the model. Drug Courts are applied widely to adult criminal cases, juvenile delinquency and truancy cases, and family court cases involving parents at risk of losing custody of their children as a result of substance use problems.

In the twenty-six years since the first Drug Court was founded in Miami/Dade County, Florida, more research has been published on the effects of Drug Courts than virtually all other criminal justice programs combined. The scientific community has put Drug Courts under a microscope and concluded that Drug Courts significantly reduce drug abuse and crime and do so at far less cost than any other justice strategy. Drug Courts improve communities by successfully getting justice-involved individuals clean and sober, stopping drug-related crime, reuniting broken families, intervening with juveniles before they embark on a debilitating life of addiction and crime, and preventing impaired driving.

This success has motivated NADCP to champion new generations of the Drug Court model, including but not limited to Veterans Treatment Courts, Reentry Courts, and Mental Health Courts. Veterans Treatment Courts link critical services and provide the structure needed for military veterans who are involved in the justice system as a result of substance abuse or mental illness to resume productive lives after combat. Reentry Courts assist individuals leaving our nation's jails and prisons to succeed on probation or parole and avoid a recurrence of drug abuse and crime. And Mental Health Courts treat and monitor those with severe and persistent mental illness who often find their way into the justice system because of their illness.

Today the award-winning NADCP is the premier national membership, training, and advocacy organization for the Drug Court model, representing over 27,000 multidisciplinary justice professionals and community leaders. NADCP hosts the largest annual training conference on drugs and crime in the nation and provides 130 training and technical assistance events each year through its professional service branches, the National Drug Court Institute, the National Center for DWI Courts, and Justice for Vets: The National Veterans Treatment Court Clearinghouse. NADCP publishes numerous scholastic and practitioner publications critical to the growth and fidelity of the Drug Court model, and works tirelessly on Capitol Hill, in the media, and in state legislatures to improve the response of the American justice system to help persons suffering from drug addiction and mental illness through effective policy, legislation, appropriations, and public education.

ACKNOWLEDGMENTS

Producing the first two volumes of the ***Adult Drug Court Best Practice Standards*** has been a tremendous undertaking, which would not have been possible but for the dedication and contributions of so many seasoned and generous professionals. This exhaustive project has been continuing for more than six years. The five standards contained in Volume II and the five preceding standards in Volume I are the products of countless hours of effort from a host of dedicated professionals who volunteered their time, energy, and intellects to the endeavor.

First, I want to thank the committee of volunteer practitioners, researchers, and subject-matter experts who developed the topics and materials contained in these standards. Second, I thank the peer reviewers who provided invaluable and meticulous feedback on each standard. Finally, I thank the NADCP Board of Directors for its leadership and vision in supporting this time- and labor-intensive effort. I reserve special recognition for Dr. Douglas Marlowe, who, as he did with Volume I, labored over every aspect of the document to ensure that it provides guidance that is clear, rooted convincingly in scientific research, and as practical as possible for Drug Court professionals.

I could not be more excited as we release this next volume. In the twenty-six years since the first Drug Court was founded, a vast body of research has proven not only that Drug Courts work, but also how they work and for whom. We now know how to structure and implement our Drug Courts to achieve the best outcomes, and it is a great privilege and honor to share this hard-won knowledge with the Drug Court field. With science guiding the hands of a compassionate and dedicated field of educated professionals, Drug Courts are poised to give hope and the gift of recovery to millions of deserving citizens, enhance public safety in our communities, and make the best use of taxpayer dollars.

Clearly the work is not done. The material contained herein must be disseminated widely to the Drug Court field and translated into day-to-day Drug Court operations. Much effort lies ahead to train practitioners on the content of the standards and put the recommended procedures into effect. In addition, future volumes of the standards will address other aspects of Drug Court procedures as new research findings become available. NADCP stands ready to deliver the requisite training, technical assistance, and knowledge development needed to enact the standards, expand the field's knowledge of best practices, and produce the best possible results for our participants and our communities.

Carolyn D. Hardin
Interim Chief Executive Officer
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**ADULT DRUG COURT
BEST PRACTICE STANDARDS**

INTRODUCTION

Until Drug Courts define appropriate standards of practice, they will be held accountable, fairly or unfairly, for the worst practices in the field. Scientists will continue to analyze the effects of weak Drug Courts alongside those of exceptional Drug Courts, thus diluting the benefits of Drug Courts. Critics will continue to tarnish the reputation of Drug Courts by attributing to them the most noxious practices of the feeblest programs. Only by defining the bounds of acceptable and exceptional practices will Drug Courts be in a position to disown poor-quality or harmful programs and set effective benchmarks for new and existing programs to achieve.

—Adult Drug Court Best Practice Standards, Volume I (NADCP, 2013; p. 1)

Volume I

In 2013, NADCP released Volume I of the *Adult Drug Court Best Practice Standards* (Standards). This landmark document was the product of more than four years of exhaustive work reviewing scientific research on best practices in substance abuse treatment and correctional rehabilitation and distilling that vast literature into measurable and enforceable practice recommendations for Drug Court professionals.

The response from the Drug Court field was immediate and profound. In the ensuing two years, twenty out of twenty-five states (80%) responding to a national survey indicated they have adopted the Standards for purposes of credentialing, funding, or training new and existing Drug Courts in their jurisdictions. The parlance of the field is literally evolving as evidence-based terminology permeates Drug Court policies and procedures. Drug Court professionals now speak routinely about targeting high-risk and high-need participants [Standard I], ensuring equivalent access and services for members of historically disadvantaged groups [Standard II], enhancing perceptions of procedural fairness during court hearings [Standard III], distinguishing proximal from distal behavioral goals and responding to participant conduct accordingly [Standard IV], and delivering evidence-based treatments matched to participants' clinical needs and prognoses for success in treatment [Standard V].

Any concerns that the Standards might sit on a shelf and collect dust vanished rapidly. Drug Courts are changing their policies and procedures in accordance with scientific findings and improving their outcomes as a result.

Volume II

Volume I marked the beginning of an ongoing process of self-evaluation and self-correction initiated by and for the Drug Court field. Before the ink dried on Volume I, NADCP launched subsequent efforts to bring Volume II to print, and those efforts have now reached fruition. Volume II picks up seamlessly where Volume I left off and describes best practices for Drug Courts on the following topics:

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VI. Complementary Treatment and Social Services. Drug Court participants often have a range of service needs extending well beyond substance abuse treatment. Standard VI addresses an array of co-occurring needs encountered frequently in Drug Courts, including best practices for delivering mental health treatment, trauma-informed services, criminal thinking interventions, family counseling, vocational or educational counseling, and prevention education to reduce health-risk behaviors.

VII. Drug and Alcohol Testing. Unless Drug Courts have accurate and timely information as to whether participants are maintaining abstinence from illicit drugs and alcohol, they have no way to apply incentives, sanctions, or treatment adjustments effectively. Standard VII describes best practices for detecting unauthorized substance use in a population that is often highly motivated and surprisingly adept at avoiding detection by standard testing methods.

VIII. Multidisciplinary Team. Recent studies have shed considerable light on the workings of the Drug Court team. Standard VIII reviews the latest research indicating which professional disciplines should be represented on the team, how team members should share information and expertise, and how often and under what circumstances team members should receive preparatory instruction and continuing-education training on Drug Court best practices.

IX. Census and Caseloads. Drug Courts need to “go to scale” and treat all eligible individuals involved in the criminal justice system. Yet studies suggest outcomes may decline if caseloads increase without ensuring that programs have sufficient resources to maintain fidelity to best practices. Standard IX identifies milestones related to the size of the Drug Court census and caseloads for supervision officers and clinicians that should trigger a reexamination of a Drug Court’s resources and adherence to best practices.

X. Monitoring and Evaluation. Drug Courts are successful in large measure because they recognized the importance of research and evaluation from their inception. Not all studies, however, employ adequate scientific methodology, thus contributing a good deal of “noise” and confusion to the scientific literature on Drug Courts. Standard X describes best practices for monitoring a Drug Court’s adherence to best practices and evaluating its impacts on substance abuse, crime, participants’ emotional health, and other important outcomes.

Procedures

NADCP employed the same procedures for developing Volume II as were employed for Volume I. The standards were drafted by a diverse and multidisciplinary committee comprising Drug Court practitioners, subject-matter experts, researchers, and state and federal policymakers. Each draft standard was peer-reviewed subsequently by at least thirty practitioners and researchers with expertise in the relevant subject matter. The peer reviewers rated the standards on the dimensions of *clarity* (what specific practices were required), *justification* (why those practices were required), and *feasibility* (how difficult it would be for Drug Courts to implement the practices). All of the standards received ratings from good to excellent and were viewed as achievable by most Drug Courts within a reasonable period of time. How long this process should take will vary from standard to standard. Drug Courts should be able to comply with some of the standards within a few months if they are not already doing so; however, other standards may require three to five years to satisfy.

None of the requirements contained in the Standards will come as a surprise to Drug Court professionals who have attended a training workshop or conference within the past five years. The research supporting these standards has been disseminated widely to the Drug Court field via conference presentations, webinars, practitioner fact sheets, and NDCI's scholarly journal, the *Drug Court Review*. Volumes I and II of the Standards are simply the first documents to compile and distill that research into concrete and measurable practice recommendations.

Future Volumes

The standards contained in Volumes I and II do not come close to addressing every practice performed in a Drug Court. Unless reliable and convincing evidence demonstrated that a practice significantly improves outcomes, it was not incorporated (yet) into a best practice standard. This should in no way be interpreted to suggest that omitted practices are unimportant or less important than the practices that were included. Practices were omitted simply because the current state of research is insufficient at this time to provide dependable guidance to the field or to impose an obligation on Drug Courts to alter their operations. Additional practices will be added to the Standards in future volumes as new studies are completed. Future standards are expected to address topics including best practices for community-supervision officers in Drug Courts; restorative-justice interventions such as community service or victim restitution; payment of fines, fees, and costs; peer and vocational mentoring; and recovery-oriented systems of care. NADCP is working actively with researchers and funders to fill these gaps in the literature and is committed to publishing related practice guidance as soon as a sufficient body of evidence is compiled.

To date, best practice standards have only been developed for Adult Drug Courts. This fact does not suggest that Adult Drug Courts are more effective or valued than other types of problem-solving courts such as Juvenile Drug Courts, DWI Courts, Family Drug Courts, or Veterans Treatment Courts. Adult Drug Courts simply have far more research on them than other types of problem-solving courts. When a sufficient body of research identifies best practices for other problem-solving court programs, NADCP will develop and release best practice standards for those programs as well.

Implementation

Putting science into practice is the greatest challenge facing the substance abuse treatment and criminal justice fields (Damschroder et al., 2009; Rudes et al., 2013; Taxman & Belenko, 2013). So far, Drug Courts are doing considerably better than most programs at following best practice standards; however, more work is needed. Programs that ignore best practices and fail to attend training conferences are the ones most likely to produce ineffective or harmful results (Carey et al., 2012; Shaffer, 2006; van Wormer, 2010) and thus to diminish the effects of Drug Courts and tarnish the reputation of the field. There is no escaping the need to redouble our efforts to disseminate best practice information widely, provide needed technical assistance to help Drug Courts bring themselves into compliance with the standards, and hold outlier programs accountable for refusing to align their practices with what works.

Responsibility for enforcing best practices is the province of state and local court and treatment systems; however, NADCP and other national organizations can and will play a critical role in

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training, consulting, and evaluating program adherence to best practices. Coordinated efforts at the state, local, and national levels will teach Drug Courts what they should be doing, why they should be doing it, and how to do it. Programs that turn a blind eye to this assistance will be readily identifiable and will ultimately face the same consequences as any other program or professional that provides deficient services below the recognized standard of care for their field.

Drug Courts have always set the highest standards for themselves. Dissatisfied with what was being done in the past, Drug Courts pushed the envelope and redesigned the criminal justice system. They brushed aside old paradigms and changed the language of justice reform. The large majority of Drug Courts can be expected to follow best practices once those practices have been identified and to save innumerable lives in the process. With a critical mass of effective programs crowding out ineffective alternatives, Drug Courts will continue to lead the way toward improved public health, public safety, and higher financial benefits for taxpayers.

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VI. COMPLEMENTARY TREATMENT AND SOCIAL SERVICES

Participants receive complementary treatment and social services¹ for conditions that co-occur with substance abuse and are likely to interfere with their compliance in Drug Court, increase criminal recidivism, or diminish treatment gains.

- A. Scope of Complementary Services
- B. Sequence and Timing of Services
- C. Clinical Case Management
- D. Housing Assistance
- E. Mental Health Treatment
- F. Trauma-Informed Services
- G. Criminal Thinking Interventions
- H. Family and Interpersonal Counseling
- I. Vocational and Educational Services
- J. Medical and Dental Treatment
- K. Prevention of Health-Risk Behaviors
- L. Overdose Prevention and Reversal

A. Scope of Complementary Services

The Drug Court provides or refers participants for treatment and social services to address conditions that are likely to interfere with their response to substance abuse treatment or other Drug Court services (*responsivity needs*), to increase criminal recidivism (*criminogenic needs*), or to diminish long-term treatment gains (*maintenance needs*). Depending on participant needs, complementary services may include housing assistance, mental health treatment, trauma-informed services, criminal-thinking interventions, family or interpersonal counseling, vocational or educational services, and medical or dental treatment. Participants receive only those services for which they have an assessed need.

B. Sequence and Timing of Services

In the first phase of Drug Court, participants receive services designed primarily to address responsivity needs such as deficient housing, mental health symptoms, and substance-related cravings, withdrawal, or anhedonia (diminished ability to experience pleasure). In the interim phases of Drug Court, participants receive services designed to resolve criminogenic needs that co-occur frequently with substance abuse, such as

¹ The term *complementary treatment and social services* refers to interventions other than substance abuse treatment that ameliorate symptoms of distress, provide for participants' basic living needs, or improve participants' long-term adaptive functioning. The term does not include restorative-justice interventions such as victim restitution, supervisory interventions such as probation home visits, or recovery-oriented services such as peer mentoring.

criminal-thinking patterns, delinquent peer interactions, and family conflict. In the later phases of Drug Court, participants receive services designed to maintain treatment gains by enhancing their long-term adaptive functioning, such as vocational or educational counseling.

C. Clinical Case Management

Participants meet individually with a clinical case manager or comparable treatment professional at least weekly during the first phase of Drug Court. The clinical case manager administers a validated assessment instrument to determine whether participants require complementary treatment or social services, provides or refers participants for indicated services, and keeps the Drug Court team apprised of participants' progress.

D. Housing Assistance

Where indicated, participants receive assistance finding safe, stable, and drug-free housing beginning in the first phase of Drug Court and continuing as necessary throughout their enrollment in the program. If professional housing services are not available to the Drug Court, clinical case managers or other staff members help participants find safe and sober housing with prosocial and drug-free relatives, friends, or other suitable persons. Participants are not excluded from participation in Drug Court because they lack a stable place of residence.

E. Mental Health Treatment

Participants are assessed using a validated instrument for major mental health disorders that co-occur frequently in Drug Courts, including major depression, bipolar disorder (manic depression), posttraumatic stress disorder (PTSD), and other major anxiety disorders. Participants suffering from mental illness receive mental health services beginning in the first phase of Drug Court and continuing as needed throughout their enrollment in the program. Mental illness and addiction are treated concurrently using an evidence-based curriculum that focuses on the mutually aggravating effects of the two conditions. Participants receive psychiatric medication based on a determination of medical necessity or medical indication by a qualified medical provider. Applicants are not denied entry to Drug Court because they are receiving a lawfully prescribed psychiatric medication [see Standard I, Target Population], and participants are not required to discontinue lawfully prescribed psychiatric medication as a condition of graduating from Drug Court [see Standard V, Substance Abuse Treatment].

F. Trauma-Informed Services

Participants are assessed using a validated instrument for trauma history, trauma-related symptoms, and posttraumatic stress disorder (PTSD). Participants with PTSD receive an evidence-based intervention that teaches them how to manage distress without resorting to substance abuse or other avoidance behaviors, desensitizes them gradually to symptoms of panic and anxiety, and encourages them to engage in productive actions that reduce the risk of retraumatization. Participants with PTSD or severe trauma-related symptoms are evaluated for their suitability for group interventions and are treated on an individual basis or in small groups when necessary to manage panic, dissociation, or

severe anxiety. Female participants receive trauma-related services in gender-specific groups. All Drug Court team members, including court personnel and other criminal justice professionals, receive formal training on delivering trauma-informed services.

G. Criminal Thinking Interventions

Participants receive an evidence-based criminal-thinking intervention after they are stabilized clinically and are no longer experiencing acute symptoms of distress such as cravings, withdrawal, or depression. Staff members are trained to administer a standardized and validated cognitive-behavioral criminal-thinking intervention such as Moral Reconation Therapy, the Thinking for a Change program, or the Reasoning & Rehabilitation program.

H. Family and Interpersonal Counseling

When feasible, at least one reliable and prosocial family member, friend, or daily acquaintance is enlisted to provide firsthand observations to staff about participants' conduct outside of the program, to help participants arrive on time for appointments, and to help participants satisfy other reporting obligations in the program. After participants are stabilized clinically, they receive an evidence-based cognitive-behavioral intervention that focuses on improving their interpersonal communication and problem-solving skills, reducing family conflicts, and eliminating associations with substance-abusing and antisocial peers and relatives.

I. Vocational and Educational Services

Participants with deficient employment or academic histories receive vocational or educational services beginning in a late phase of Drug Court. Vocational or educational services are delivered after participants have found safe and stable housing, their substance abuse and mental health symptoms have resolved substantially, they have completed a criminal-thinking intervention, and they are spending most or all of their time interacting with prosocial and sober peers. Vocational interventions are standardized and cognitive-behavioral in orientation and teach participants to find a job, keep a job, and earn a better or higher-paying job in the future through continuous self-improvement. Participants are required to have a stable job, be enrolled in a vocational or educational program, or be engaged in comparable prosocial activity as a condition of graduating from Drug Court. Continued involvement in work, education, or comparable prosocial activity is a component of each participant's continuing-care plan.

J. Medical and Dental Treatment

Participants receive immediate medical or dental treatment for conditions that are life-threatening, cause serious pain or discomfort, or may lead to long-term disability or impairment. Treatment for nonessential or nonacute conditions that are exacerbated by substance abuse may be provided in a late phase of Drug Court or included in the participant's continuing-care plan.

K. Prevention of Health-Risk Behaviors

Participants complete a brief evidence-based educational curriculum describing concrete measures they can take to reduce their exposure to sexually transmitted and other communicable diseases.

L. Overdose Prevention and Reversal

Participants complete a brief evidence-based educational curriculum describing concrete measures they can take to prevent or reverse drug overdose.

COMMENTARY**A. Scope of Complementary Services**

Drug Court participants frequently have needs for treatment and social services that extend well beyond substance abuse treatment. National and statewide studies have found that substantial proportions of Drug Court participants suffered from a serious co-occurring mental health or medical disorder, were chronically unemployed, had low educational achievement, were homeless, or had experienced physical or sexual abuse or other trauma (see Table 1).

TABLE 1	COMPLEMENTARY NEEDS IDENTIFIED IN NATIONAL AND STATEWIDE STUDIES OF DRUG COURTS
Complementary Need	Percentage of Participants
Any mental health problem/disorder	63%
Major depression	16%–39%
Posttraumatic stress disorder (PTSD)	10%
Anxiety disorder other than PTSD	9%
Bipolar disorder	8%
Chronic medical condition	26%
Unemployed	54%–72%
Less than a high school diploma or GED	32%–38%
Homeless	11%–47%
Abuse or trauma history	27%–29%

Sources: Cissner et al. (2013); Green & Rempel (2012); Peters et al. (2012).

Drug Courts are more effective and cost-effective when they offer complementary treatment and social services to address these co-occurring needs. A multisite study of approximately seventy Drug Courts found that programs were significantly more effective at reducing crime when they offered mental health treatment, family counseling, and parenting classes and were marginally more effective when they offered medical and dental services (Carey et al., 2012). The same study determined that Drug Courts were more cost-effective when they helped participants find a job, enroll in an educational program, or obtain sober and supportive housing. Similarly, a statewide study of eighty-six Drug Courts in New York found that programs were significantly more effective at reducing crime when they assessed participants for trauma and other mental health treatment needs, and delivered mental health, medical, vocational, or educational services where indicated (Cissner et al., 2013).

Studies do not, however, support a practice of delivering the same complementary services to all participants. Drug Courts that required all participants to receive educational or employment services were

determined in one meta-analysis to be less effective at reducing crime than Drug Courts that matched these services to the assessed needs of the participants (Shaffer, 2006). Requiring participants to receive unnecessary services wastes time and resources and can make outcomes worse by placing excessive demands on participants and interfering with the time they have available to engage in productive activities (Gutierrez & Bourgon, 2012; Lowenkamp et al., 2006; Prendergast et al., 2013; Smith et al., 2009; Vieira et al., 2009; Viglione et al., 2015). Evidence also suggests participants may become resentful, despondent, or anxious if they are sanctioned for failing to meet excessive or unwarranted demands, a phenomenon referred to as learned helplessness or ratio burden (Seligman, 1975). Under such circumstances, behavior fails to improve, and participants may leave treatment prematurely (Marlowe & Wong, 2008). If a Drug Court team cannot articulate a sound rationale for requiring a participant to receive a given service, then the team should reconsider requiring that service.

B. Sequence and Timing of Services

Timing is critical to the successful delivery of complementary treatment and social services. Outcomes are significantly better when rehabilitation programs address complementary needs in a specific sequence. This finding has important implications for designing the phase structure in a Drug Court. The first phase of Drug Court should focus primarily on resolving conditions that are likely to interfere with retention or compliance in treatment (responsivity needs). This process may include meeting participants' basic housing needs, stabilizing mental health symptoms if present, and ameliorating acute psychological or physiological symptoms of addiction, such as cravings, anhedonia, or withdrawal. Subsequently, the interim phases of Drug Court should focus on resolving needs that increase the likelihood of criminal recidivism and substance abuse (criminogenic needs). This process includes initiating sustained abstinence from drugs and alcohol, addressing dysfunctional or antisocial thought patterns, eliminating delinquent peer associations, and reducing family conflict. Finally, later phases of Drug Court should address remaining needs that are likely to undermine the maintenance of treatment gains (maintenance needs). This process may include providing vocational or educational assistance, parent training, or other interventions designed to enhance participants' activities of daily living (ADL) skills.²

Responsivity Needs. When participants first enter Drug Court, one of the most pressing goals is to ensure that they remain in treatment and comply with other reporting obligations. This objective requires Drug Courts to resolve symptoms or conditions that are likely to interfere with attendance or engagement in treatment. Such conditions are commonly referred to as responsivity needs because they interfere with a person's response to rehabilitation efforts (Andrews & Bonta, 2010; Smith et al., 2009). Although responsivity needs do not necessarily cause or exacerbate crime, they nevertheless must be addressed early in treatment to prevent participants from failing or dropping out of treatment prematurely (Hubbard & Pealer, 2009; Karno & Longabaugh, 2007).

Responsivity needs that are commonly encountered in Drug Courts include severe mental illness and homelessness or unstable housing (Cissner et al., 2013; Green & Rempel, 2012; Peters et al., 2012). Although these conditions usually do not cause crime (Andrews & Bonta, 2010; Bonta et al., 1998; Gendreau et al., 1996), they have a marked tendency to undermine the effectiveness of Drug Courts and other correctional rehabilitation programs (Gray & Saum, 2005; Hickert et al., 2009; Johnson et al., 2011; Mendoza et al., 2013; Young & Belenko, 2002). To avoid premature termination from Drug Court, these responsivity needs must be addressed, when present, beginning in the first phase of treatment and continuing as needed throughout participants' enrollment in the program.

Criminogenic Needs. Criminogenic needs refer to disorders or conditions that cause or exacerbate crime (Andrews & Bonta, 2010). Drug and alcohol dependence are highly criminogenic needs (Bennett et al., 2008; Walters, 2015), which explains why they are the primary focus of most interventions in Drug Courts. Other criminogenic needs that are encountered frequently in Drug Courts include criminal-thinking

² This phase structure assumes a Drug Court is serving high-risk and high-need participants [see Standard I]. If a Drug Court serves individuals who are not addicted to drugs or alcohol or suffering from a serious mental illness, it may be advisable to deliver vocational, educational or other maintenance interventions beginning in an early phase of the program (Cresswell & Deschenes, 2001; Gallagher, 2013a; Vito & Tewksbury, 1998).

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patterns, impulsivity, family conflict, and delinquent peer affiliations (Green & Rempel, 2012; Hickert et al., 2009; Jones et al., 2015).

Studies have reported improved outcomes when Drug Courts provided services to address these criminogenic needs. For example, superior outcomes have been reported when Drug Court participants learned to apply effective and prosocial decision-making skills, such as learning to think before they act, to consider the potential consequences of their actions, and to recognize their own role in interpersonal conflicts (Cheesman & Kunkel, 2012; Heck, 2008; Kirchner & Goodman, 2007; Lowenkamp et al., 2009; Vito & Tewksbury, 1998). Similarly, studies found that crime and substance abuse declined significantly when Drug Court participants spent less time interacting with delinquent peers, spent more time interacting with prosocial peers and relatives, and reported fewer conflicts with family members (Green & Rempel, 2012; Hickert et al., 2009; Shaeffer et al., 2010; Wooditch et al., 2013).

Maintenance Needs. Some needs, such as poor job skills, illiteracy, or low self-esteem, are often the result of living a nonproductive or antisocial lifestyle rather than the cause of that lifestyle (Hickert et al., 2009; Wooditch et al., 2013). Treating such noncriminogenic needs before one treats criminogenic needs is associated with increased criminal recidivism, treatment failure, and other undesirable outcomes (Andrews & Bonta, 2010; Andrews et al., 1990; Smith et al., 2009; Vieira et al., 2009). Nevertheless, if these needs are ignored over the long term, they are likely to interfere with the maintenance of treatment gains. Improvements in certain maintenance needs, such as improved educational achievement or job skills, predict better long-term persistence of treatment effects (Leukefeld et al., 2007).

The important point is that improvements in maintenance needs rarely occur until after the more pressing responsivity and criminogenic needs have been resolved. Participants are unlikely, for example, to improve their job performance until after they have stopped experiencing debilitating symptoms of addiction or mental illness, stopped associating with delinquent peers, and relinquished self-centered attitudes and impulsive behaviors (Guastafarro, 2012; Samenow, 2014). After participants are stabilized clinically and have achieved a reasonable period of sobriety, maintenance services designed to enhance their adaptive functioning and ADL skills help to ensure the gains are sustained. Outcomes are also significantly better when continued involvement in maintenance activities after discharge is a requirement for graduation and a component of each participant's continuing-care plan (Carey et al., 2012).

C. Clinical Case Management

Studies consistently find that Drug Courts are more effective and cost-effective when participants meet individually with a clinical case manager or comparable treatment professional at least weekly during the first phase of the program (Carey et al., 2012; Cissner et al., 2013; Zweig et al., 2012). As described previously, Drug Courts must identify a range of complementary needs among participants, refer participants for indicated services, and ensure the services are delivered in an effective sequence. To do otherwise risks wasting resources and making outcomes worse for some participants. These complicated tasks require input from a professionally trained clinical case manager or clinician who is competent to perform clinical and social service assessments, understands how services should be sequenced and matched to participant needs, and is skilled at monitoring and reporting on participant progress (Monchick et al., 2006; Rodriguez, 2011).

Typically, clinical case managers are addiction counselors, social workers, or psychologists who have received specialized training to assess participant needs, broker referrals for indicated services, coordinate care between partner agencies, and report progress information to other interested professionals (Monchick et al., 2006; Rodriguez, 2011). In some Drug Courts, probation officers or other criminal justice professionals may serve as court case managers, to be distinguished from clinical case managers. Typically, court case managers administer brief screening instruments designed to identify participants requiring more in-depth clinical assessments. Participants scoring above established thresholds on the screening instruments are referred for further evaluation by a clinically trained treatment professional.

Broadly speaking, there are four basic models of clinical case management (Hesse et al., 2007; Rapp et al., 2014):

- *Brokerage Model*—The least intensive form of case management, the brokerage model involves assessing participants and linking them to indicated services.
- *Generalist or Clinician Model*—In the most common form of case management, the Generalist case manager assesses participant needs and delivers some or all of the indicated services.
- *Assertive Community Treatment (ACT) Model*—The most intensive form of case management, the ACT Model provides around-the-clock access to a multidisciplinary team of professionals that delivers wrap-around services in the community designed to meet an array of treatment and social-service needs.
- *Strengths-Based Model*—A strengths-based philosophy may be applied in the context of any of the above models. It focuses on leveraging participants' natural resources and encouraging participants to take an active role in setting treatment goals and selecting treatment options.

Meta-analyses reveal that all four case management models significantly increase referrals for indicated services and retain participants longer in treatment; however, they have relatively small effects on substance abuse, crime, and other long-term outcomes (Hesse et al., 2007; Rapp et al., 2014). Whether a program produces long-term improvements depends ultimately on the quality and quantity of treatment and social services that are delivered. No evidence suggests any one case management model is superior to another; however, the models were developed for different types of programs serving individuals with different clinical and social service profiles. The generalist model was developed primarily for use in outpatient treatment settings where a primary therapist commonly delivers or coordinates the delivery of various components of a participant's care. Although few Drug Court studies have provided a clear description of the case management services that were provided, the generalist model appears to be used most frequently in adult Drug Courts (Carey et al., 2012; Cissner et al., 2013; Zweig et al., 2012).

The brokerage model was developed for participants who are served by more than one agency or system. For example, some substance abuse treatment programs may lack the required expertise to deliver mental health treatment or vocational rehabilitation. As a result, participants must be referred to another agency for a portion of their care. A clinical case manager is required to broker the referral, reconcile conflicting demands that may be placed on participants by different agencies, and report on participant progress to the Drug Court team.

A specific model of case management, called Treatment Accountability for Safer Communities or Treatment Alternatives to Street Crime (TASC), was designed to bridge gaps between the substance abuse, mental health, and criminal justice systems. TASC programs typically apply a brokerage or generalist model depending on whether treatment is available within the criminal justice system or must be brokered through another system or agency. Evidence is convincing that TASC programs increase participants' access to services and retention in treatment; however, impacts on substance abuse and crime have been mixed (Anglin et al., 1999; Ventura & Lambert, 2004). As was already noted, the key to successful outcomes depends on the quality and quantity of treatment and social services that are delivered (Clark et al., 2013; Cook, 2002; Rodriguez, 2011). Outcomes are more consistently favorable when TASC case management is delivered in conjunction with intensive evidence-based treatment as in Drug Courts (Monchick et al., 2006). Therefore, training on the TASC model or a comparable case management model is important for staff members providing clinical case management services in Drug Courts.

Finally, the ACT model was developed for use with seriously impaired individuals who have a wide range of mental health and social service needs (McLellan et al., 1998, 1999). This intensive model of case management has been applied successfully in the context of a mental health court (Braude, 2005) and a community court serving persons with serious and persistent mental illness or social service needs (Somers et al., 2014). Training on the ACT model of case management is advisable for Drug Courts serving seriously impaired individuals suffering from co-occurring mental illness, chronic homelessness, or other severe functional impairments.

Regardless of which model of case management is applied, outcomes are superior when case managers administer reliable and valid needs-assessment instruments (Andrews & Bonta, 2010; Andrews et al., 2006). [Appendix C provides examples of validated instruments designed to assess clinical and

criminogenic needs among persons in substance abuse treatment and the criminal justice system.] Whether needs assessments should be administered repeatedly during the course of treatment is an open question. Although evidence suggests changes in need scores correlate with progress in treatment (Greiner et al., 2015; Serin et al., 2013; Vose et al., 2013; Wooditch et al., 2013), little guidance is available to determine when or how to alter treatment conditions in light of changing scores (Serin et al., 2013). Until such guidance is available, Drug Courts are advised to rely on objective indices of participant progress, such as drug test results and treatment attendance rates, to make decisions about adjusting treatment and social services.

On a final note, a critical function of case management is linking participants to public benefits and other subsidies to which they are legally entitled. For example, under the Affordable Care Act (ACA), Drug Court participants may be eligible for medical or mental health care benefits pursuant to Medicaid expansion or newly created health-insurance exchanges (Frescoln, 2014). Court case managers or clinical case managers must leverage these financial resources and enroll participants for eligible benefits to meet participants' needs for substance abuse treatment and other complementary services.

D. Housing Assistance

Participants are unlikely to succeed in treatment if they do not have a safe, stable, and drug-free place to live (Morse et al., 2015; Quirouette et al., 2015). No study was identified that has examined the impact of housing assistance on Drug Court outcomes. However, studies in similar contexts have reported improved outcomes when housing assistance was provided for parolees reentering the community after prison (Clark, 2014; Lutze et al., 2014), in community courts for persons suffering from serious and persistent mental illness (Kilmer & Sussell, 2014; Lee et al., 2013), and in programs serving homeless military veterans (Elbogen et al., 2013; Winn et al., 2014).

Some Drug Courts may have a policy of denying entry to persons who do not have a stable place of residence. Such a policy is likely to have the unintended effect of excluding the highest-risk and highest-need individuals—those who need Drug Court the most—from participation in Drug Court (Morse et al., 2015; Quirouette et al., 2015). The preferable course of action is to provide housing assistance, where indicated, beginning in the first phase of Drug Court and continuing as needed throughout participants' enrollment in the program. If professional housing services are not available to a Drug Court, then clinical case managers or other staff members should make every effort to help participants find safe and stable housing with prosocial and drug-free relatives, friends, or other suitable individuals.

E. Mental Health Treatment

Approximately two-thirds of Drug Court participants report serious mental health symptoms and roughly one-quarter have a diagnosed Axis I psychiatric disorder, most commonly major depression, bipolar disorder, PTSD, or other anxiety disorder (Cissner et al., 2013; Green & Rempel, 2012; Peters et al., 2012). Mental illness, by itself, is ordinarily not a criminogenic need (Bonta et al., 1998; Elbogen & Johnson, 2009; Gendreau et al., 1996; Peterson et al., 2014; Phillips et al., 2005; Prins et al., 2014); however, it is a responsivity need that can interfere significantly with the effectiveness of Drug Courts and other rehabilitation programs (Gray & Saum, 2005; Hickert et al., 2009; Johnson et al., 2011; Manchak et al., 2014; Mendoza et al., 2013; Ritsher et al., 2002; Young & Belenko, 2002). Moreover, when mental illness is combined with substance abuse, the odds of recidivism increase significantly—although the magnitude of this effect is smaller than for most other criminogenic risk factors, such as a participant's criminal history or association with delinquent peers (Andrews & Bonta, 2010; Peters et al., 2015; Rezanoff et al., 2013).

Mental illness and substance abuse may co-occur in a given case for several reasons. Substance abuse may trigger or exacerbate mental illness, mentally ill individuals may abuse substances in a misguided effort to self-medicate psychiatric symptoms, or the two disorders may emerge independently in a person who has a generalized vulnerability to stress-related illness (Ross, 2008). Causality aside, treating either disorder alone without treating both disorders simultaneously is rarely, if ever, successful. Addiction and mental illness are reciprocally aggravating conditions, meaning that continued symptoms of one disorder are likely to precipitate relapse in the other disorder (Chandler et al., 2004; Drake et al., 2008). For example, a

formerly depressed person who continues to abuse drugs is likely to experience a resurgence of depressive symptoms. Conversely, a person recovering from addiction who continues to suffer from depression is at risk for relapsing to drug abuse. For this reason, best practice standards for Drug Courts and other treatment programs require mental illness and addiction to be treated concurrently as opposed to consecutively (Drake et al., 2004; Kushner et al., 2014; Mueser et al., 2003; Osher et al., 2012; Peters, 2008; Steadman et al., 2013). Whenever possible, both disorders should be treated in the same facility by the same professional(s) using an integrated treatment model that focuses on the mutually aggravating effects of the two conditions. The Substance Abuse and Mental Health Services Administration (SAMHSA, 2010) has published therapist toolkits to assist in delivering evidence-based integrated treatments for co-occurring substance-use and mental health disorders.

Participants should also have unhindered access to medical providers qualified to prescribe and monitor response to psychiatric medications (Kushner et al., 2014; Steadman et al., 2013). In one study, Drug Court participants who were prescribed psychiatric medications were seven times more likely to graduate successfully from the program than participants with psychiatric symptoms who did not receive psychiatric medications (Gray & Saum, 2005). Thus, for Drug Courts to deny participants access to psychiatric medication or require them to discontinue legally prescribed psychiatric medication as a condition of entering or graduating from Drug Court is not appropriate [see also Standard I, Target Population, and Standard V, Substance Abuse Treatment]. A participant should only be denied psychiatric medication if the decision is based on expert medical evidence from a qualified physician who has examined the participant and is adequately informed about the facts of the case (Peters & Osher, 2004; Steadman et al., 2013).

F. Trauma-Informed Services

More than one-quarter of Drug Court participants report having been physically or sexually abused in their lifetime or having experienced another serious traumatic event, such as a life-threatening car accident or work-related injury (Cissner et al., 2013; Green & Rempel, 2012). Among female Drug Court participants, studies have found that more than 80% experienced a serious traumatic event in their lifetime, more than half were in need of trauma-related services, and over a third met diagnostic criteria for PTSD (Messina et al., 2012; Powell et al., 2012; Sartor et al., 2012).

Unlike most types of mental illness which are typically noncriminogenic, individuals in the criminal justice system who have PTSD are approximately one and a half times more likely to reoffend than those without PTSD (Sadeh & McNiel, 2015). Moreover, as is true for many forms of mental illness, individuals with PTSD are significantly more likely to drop out or to be discharged prematurely from substance abuse treatment than individuals without PTSD (Mills et al., 2012; Read et al., 2004; Saladin et al., 2014). For these reasons, addressing trauma-related symptoms beginning in the first phase of Drug Court and continuing as necessary throughout participants' enrollment in the program is essential.

Most research on treatment of PTSD and other trauma-related syndromes has been conducted with military veterans or women in gender-specific treatment programs. For persons suffering from a diagnosed PTSD, evidence-based treatments are manualized, standardized, and cognitive-behavioral in orientation (Benish et al., 2008). Effective interventions focus on the following objectives (Benish et al., 2008; Bisson et al., 2007; Bradley et al., 2005; Mills et al., 2012):

- Creating a safe and dependable therapeutic relationship between the participant and therapist
- Helping participants deal with anger, anxiety, and other negative emotions without lashing out or engaging in avoidance behaviors such as substance abuse
- Assisting participants to construct a coherent “narrative” or understanding of the traumatic events that points toward productive actions (For example, many trauma victims believe they were to blame for past traumas or are helpless to prevent future traumas. Helping participants absolve themselves of guilt for past events and learn effective behavioral strategies to avoid future retraumatization is far more productive.)
- Exposing participants, in tolerable dosages, to memories or images of the event in a manner that gradually desensitizes them to associated feelings of panic and anxiety

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Web sites providing additional information about evidence-based treatments for PTSD are listed in Appendix D.

In a randomized controlled experiment, female Drug Court participants with trauma histories who received manualized cognitive-behavioral PTSD treatments—Helping Women Recover (Covington, 2008) or Beyond Trauma (Covington, 2003)—in gender-specific groups were more likely to graduate from Drug Court, were less likely to receive a jail sanction in the program, and reported more than twice the reduction in PTSD symptoms than participants with trauma histories who did not receive PTSD treatment (Messina et al., 2012). In another study, female Drug Court participants who received similar interventions—trauma-focused cognitive-behavioral therapy or abuse-focused cognitive-behavioral therapy—reported substantial reductions in substance use and mental health symptoms as well as improvements in housing and employment (Powell et al., 2012). Given the design of these studies, separating the effects of the PTSD treatments from the effects of the gender-specific groups is not possible. Studies have reported superior outcomes when women in the criminal justice system received various types of substance abuse treatment in female-only groups (Grella, 2008; Kissin et al., 2013; Liang & Long, 2013; Morse et al., 2013). Given the current state of knowledge, the best practice is to deliver trauma-related services for women in female-only groups because this combination of services clearly enhances outcomes for these participants.

Not all individuals who experience trauma will develop PTSD or require PTSD treatment, nor can Drug Courts assume that past trauma was the cause of a participant's substance abuse problem or criminal history (Saladin et al., 2014). In some cases, trauma is the result rather than the cause of a participant's substance abuse problem or criminal involvement. Persons who engage in substance abuse or crime often expose themselves repeatedly to the potential for trauma; therefore, treating trauma symptoms without paying equivalent attention to substance abuse and other criminogenic needs is unlikely to produce sustainable improvements.

Although some participants with trauma histories do not require formal PTSD treatment, all staff members, including court personnel and other criminal justice professionals, need to be *trauma-informed* for all participants (Bath, 2008). Staff members should remain cognizant of how their actions may be perceived by persons who have serious problems with trust, are paranoid or unduly suspicious of others' motives, or have been betrayed, sometimes repeatedly, by important persons in their lives. Safety, predictability and reliability are critical for treating such individuals. Several practice recommendations should be borne in mind (Bath, 2008; Covington, 2003; Elliott et al., 2005; Liang & Long, 2013):

- Staff members should strive continually to avoid inadvertently retraumatizing participants. For example, responding angrily to participant infractions, ignoring participants' fears or concerns, maintaining a chaotic or noisy group-counseling environment, or performing urine drug testing in a public or disrespectful manner may reawaken feelings of shame, fear, guilt, or panic in formerly traumatized individuals.
- Staff should remain true to their word, including following policies and procedures as described in the program manual and applying incentives and sanctions as agreed. Too much flexibility, no matter how well-intentioned, may seem unfair and unpredictable to persons who have fallen victim to unexpected dangers in the past.
- Staff should provide clear instructions in advance to participants concerning behaviors that are expected and prohibited in the program. Individuals with trauma histories need to understand the rules and to be prepared for what will occur in the event of an accomplishment or infraction.
- Staff should start and end counseling sessions, court hearings, and other appointments on time, at the agreed-upon location, and according to an agreed-upon structure and format. If participants cannot rely on staff to follow a basic itinerary, relying on those same staff persons for trustworthy support, feedback, and counseling may prove difficult for participants.
- Participants with PTSD or severe trauma-related symptoms, such as panic or dissociation (feeling detached from one's surroundings), may not be suitable candidates for group interventions, especially in the early stages of treatment (Yalom & Leszcz, 2005). Such individuals may need to be treated on an individual basis or in small groups with carefully selected group members who are nonthreatening

and nonpredatory. As was noted earlier, female participants with trauma histories are especially well suited for gender-specific groups (Liang & Long, 2013; Messina et al., 2012).

- Participants with histories of childhood-onset abuse or neglect may be at risk for developing a severe personality disorder such as borderline personality disorder. These individuals may have considerable difficulty trusting others, controlling overwhelming feelings of anger or depression, and containing their impulses. Manualized cognitive-behavioral treatments, such as dialectical behavior therapy (Linehan, 1996), have been shown to improve outcomes in these difficult cases (Dimeff & Koerner, 2007; Linehan et al., 1999). These complicated treatments require specialized training and continuous supervision to help staff deal with uncomfortable and confusing reactions that are commonly engendered in these challenging cases.

G. Criminal Thinking Interventions

As stated earlier, criminal-thinking patterns are observed frequently among Drug Court participants (Jones et al., 2015) and may contribute to program failure (responsivity need) and criminal recidivism (criminogenic need) (Gendreau et al., 1996; Helmond et al., 2015; Knight et al., 2006; Walters, 2003). Some Drug Court participants have considerable difficulty seeing other people's perspectives, recognizing their role in interpersonal conflicts, or anticipating consequences before they act. Moreover, they may hold counterproductive attitudes or values, such as assuming that all people are untrustworthy and motivated to manipulate or dominate others. Given such antisocial sentiments, these participants are often viewed as suspicious or manipulative in character, get into repeated conflicts with others, and fail to learn from negative social interactions.

Several manualized cognitive-behavioral interventions address criminal-thinking patterns among individuals addicted to drugs or charged with crimes. Evidence-based curricula demonstrating improved outcomes in Drug Courts and similar programs include but are not limited to Moral Reconation Therapy (Cheesman & Kunkel, 2012; Heck, 2008; Kirchner & Goodman, 2007), Thinking for a Change (Lowenkamp et al., 2009), and Reasoning & Rehabilitation (Cullen et al., 2012; Tong & Farrington, 2006). Other curricula focused specifically on the needs of men in the criminal justice system, such as Habilitation, Empowerment and Accountability Therapy (Turpin & Wheeler, 2012; Vito & Tewksbury, 1998) and Helping Men Recover (Covington et al., 2011), are undergoing development and effectiveness testing in Drug Courts. Additional information about evidence-based criminal-thinking interventions is provided in Appendix D.

Studies have not determined when delivering criminal-thinking interventions is most beneficial. Clinical experience suggests the most beneficial time to introduce these interventions is after participants are stabilized in treatment and no longer experiencing acutely debilitating symptoms such as cravings, withdrawal, or anhedonia (Milkman & Wanberg, 2007). Until participants are no longer in acute distress, expecting them to benefit from a cognitive-behavioral intervention that requires them to maintain consistent attention and cognitive endurance is unrealistic. Participants should be stabilized clinically before a Drug Court can reasonably expect them to think flexibly about the motivations for their behaviors and the potential ramifications of continuing in their current behavioral patterns.

H. Family and Interpersonal Counseling

Reductions in substance abuse and crime go hand in hand with reduced family conflict, fewer interactions with delinquent relatives and peers, and increased interactions with sober and prosocial individuals (Berg & Huebner, 2011; Fergusson et al., 2002; Knight & Simpson, 1996; Wooditch et al., 2013; Wright & Cullen, 2004). These findings hold true in Drug Courts as they do in most correctional rehabilitation programs (Green & Rempel, 2012; Hickert et al., 2009).

Most studies of family treatments in Drug Courts have been conducted in the context of Family Drug Courts or Juvenile Drug Courts. Results have demonstrated consistently superior outcomes when manualized, cognitive-behavioral family interventions were added to the Drug Court curriculum, including Strengthening Families and Celebrating Families! (Brook et al., 2015) and modified versions of multidimensional family therapy (Dakof et al., 2009, 2010, 2015), multisystemic therapy (Henggeler et al., 2006), and functional family therapy (Datchi & Sexton, 2013). [Further information about these and other

evidence-based family treatments is provided in Appendix D.] Each of these treatments focuses on lessening familial conflict, reducing interactions with drug-using and antisocial peers and relatives, improving communication skills, and enhancing problem-solving skills. In the beginning of treatment, prosocial and drug-free family members, friends, or daily acquaintances are trained by staff to monitor participant behavior reliably, reinforce prosocial activities, respond appropriately and helpfully to problematic behaviors, reduce tension and conflict, and deescalate confrontations. As therapy progresses, treatment focuses on teaching all parties effective communication and problem-solving skills.

Studies have not determined when delivering family or interpersonal counseling in Drug Courts is most beneficial. Given the powerful association between family functioning and criminal justice outcomes, these services should be delivered as soon as practicable. Outcomes in substance abuse treatment are significantly better when at least one reliable and prosocial family member, friend, or close acquaintance is enlisted early in treatment to help the participant arrive on time for appointments and comply with other obligations in the program, such as following a curfew, adhering to prescribed medications, and avoiding forbidden locations like bars (Meyers et al., 1998; Roozen et al., 2010). The same individual may be enlisted to provide helpful observations to staff about the participant's conduct outside of treatment (Kirby et al., 1999). After participants are stabilized clinically, family interventions should focus on improving communication skills, altering maladaptive interactions, reinforcing prosocial behaviors, and reducing interpersonal conflicts.

I. Vocational and Educational Services

Approximately one-half to three-quarters of Drug Court participants have poor work histories or low educational achievement (Cissner et al., 2013; Deschenes et al., 2009; Green & Rempel, 2012; Hickert et al., 2009; Leukefeld et al., 2007). Being unemployed or having less than a high school diploma or general educational development (GED) certificate predicts poor outcomes in Drug Courts (DeVall & Lanier, 2012; Gallagher, 2013b; Gallagher et al., 2015; Mateyoke-Scrivener et al., 2004; Peters et al., 1999; Roll et al., 2005; Shannon et al., 2015) as it does in most other substance abuse treatment (Keefer, 2013) and correctional rehabilitation programs (Berg & Huebner, 2011; Wright & Cullen, 2004).

Unfortunately, few vocational or educational interventions have been successful at reducing crime (Aos et al., 2006; Cook et al., 2014; Farabee et al., 2014; Wilson et al., 2000) or substance abuse (Lidz et al., 2004; Magura et al., 2004; Platt, 1995). Disappointing results have commonly been attributable to poor quality and timing of the interventions. Many vocational programs amount to little more than job-placement services, which alert participants to job openings, place them in a job, or help them conduct a job search. Placing high-risk and high-need individuals in a job is unlikely to be successful if they continue to crave drugs or alcohol, experience serious mental health symptoms, associate with delinquent peers, or respond angrily or impulsively when they are criticized or receive negative feedback from others (Coviello et al., 2004; Lidz et al., 2004; Magura et al., 2004; Platt, 1995; Samenow, 2014). Improvements in education and employment rarely occur until after participants are stabilized clinically, cease interacting with delinquent peers, and learn to deal with frustration in a reasonably effective and mature manner.

At least two studies in Drug Courts have reported improved outcomes when unemployed or underemployed participants received a manualized, cognitive-behavioral vocational intervention. The effective interventions taught participants not only how to find a job, but also how to keep the job by behaving responsibly and dependably and how to land a better or higher-paying job in the future by continually honing their skills and productivity (Deschenes et al., 2009; Leukefeld et al., 2007). Comparable studies in drug abuse treatment reported improved outcomes when participants learned to interact effectively with coworkers and employers and resolve interpersonal conflicts on the job (Platt et al., 1993; Platt, 1995).

Studies have not determined when administering vocational or educational interventions is most beneficial. For high-risk and high-need individuals, these services are best introduced late in the course of Drug Court after participants have secured safe and stable housing, their addiction and mental health symptoms have resolved substantially, they have completed a criminal-thinking intervention, and they are spending most or all of their time interacting with prosocial, sober, and supportive peers (Magura et al., 2004; Platt, 1995). For many high-risk and high-need participants, this preparatory process may require at least six months of

treatment, and twelve months may be needed for individuals with serious substance use disorders or mental illness (Gottfredson et al., 2007; Peters et al., 2002).

J. Medical and Dental Treatment

Approximately one-quarter of Drug Court participants suffer from chronic medical or dental conditions that cause them serious discomfort, require ongoing medical attention, or interfere with their daily functioning (Green & Rempel, 2012). Medical and dental problems are typically maintenance needs, meaning they are most often a result rather than the cause of substance abuse and crime but can interfere with the maintenance of treatment gains. (An obvious exception is participants who become addicted to prescription medications during the course of medical or dental treatment.) Evidence suggests providing medical or dental treatment can improve outcomes for some Drug Court participants (Carey et al., 2012). Moreover, for humanitarian reasons, treating pain or discomfort regardless of the impact on criminal justice outcomes is always important.

No study has determined when addressing medical or dental concerns in Drug Courts is most appropriate. Needless to say, conditions that are life-threatening or may cause long-term disability should be treated immediately. However, waiting until later phases of Drug Court to treat nonessential or nonacute conditions that are exacerbated or maintained by substance abuse may be prudent. Outcomes may be better if medical or dental services are delivered after participants have achieved sobriety and relinquished other antisocial behaviors. For example, participants who abuse methamphetamine often have serious dental problems (American Dental Association, n.d.). If these dental problems are not causing acute distress, it might be appropriate to wait until the participant has stopped using methamphetamine before attempting dental repairs. Continued substance abuse risks undoing dental efforts and may cause a participant to discontinue dental treatment prematurely. A more efficient use of resources may be to address nonessential dental or medical treatment in a late phase of Drug Court or as part of a participant's continuing-care plan so as to maintain and extend the Drug Court's beneficial effects. A logical first step is to refer participants for routine medical and dental checkups to establish relationships with health care providers and begin a long-term process of preventive and routine medical and dental care.

K. Prevention of Health-Risk Behaviors

Alarmingly high percentages of Drug Court participants engage in behaviors which put them at serious risk for contracting human immunodeficiency virus (HIV) and other sexually transmitted diseases (STDs). In some studies, approximately 50% to 85% of Drug Court participants reported engaging in frequent unprotected sex with multiple sex partners (Festinger et al., 2012; Robertson et al., 2012; Tolou-Shams et al., 2012). Drug Court participants were found in one study to lack basic knowledge about simple self-protective measures they can take to reduce their health-risk exposure, such as using condoms and cleaning injection needles (Robertson et al., 2012).

A recent systematic review identified several brief educational interventions that are proven to reduce HIV risk behaviors among drug-addicted persons in the criminal justice system (Underhill et al., 2014). [Additional resources for identifying effective health-risk prevention programs are provided in Appendix D.] Most effective interventions are brief and inexpensive to administer, and some can be delivered via computer or videotape with minimal burden on staff. The criminal justice system is a major vector for the spread of HIV, STDs, and other serious communicable diseases (Belenko et al., 2004; Spaulding et al., 2009). Impacts on crime and substance abuse aside, Drug Courts have a responsibility to reduce the chances that participants will contract a life-threatening or incurable illness, especially in light of the fact that effective interventions can be delivered at minimal cost and burden to the program.

L. Overdose Prevention and Reversal

Unintentional overdose deaths from illicit and prescribed opiates have more than tripled in the past fifteen years (Meyer et al., 2014). Individuals addicted to opiates are at especially high risk for overdose death following release from jail or prison because tolerance to opiates decreases substantially during periods of incarceration (Dolan et al., 2005; Strang, 2015; Strang et al., 2014).

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Drug Courts should educate participants, their family members, and close acquaintances about simple precautions they can take to avoid or reverse a life-threatening drug overdose. At a minimum, this should include providing emergency phone numbers and other contact information to use in the event of an overdose or similar medical emergency.

As permitted by law, Drug Courts should also support local efforts to train Drug Court personnel, probation officers, law enforcement, and other persons likely to be first responders to an overdose on the safe and effective administration of overdose-reversal medications such as naloxone hydrochloride (naloxone or Narcan). Naloxone is nonaddictive, nonintoxicating, poses a minimal risk of medical side effects, and can be administered intranasally by nonmedically trained laypersons (Barton et al., 2002; Kim et al., 2009). The Centers for Disease Control and Prevention (2012) estimates that more than 10,000 potentially fatal opiate overdoses have been reversed by naloxone administered by nonmedical laypersons. Studies in the U.S. and Scotland confirm that educating at-risk persons and their significant others about ways to prevent or reverse overdose, including the use of naloxone, significantly reduces overdose deaths (National Institute on Drug Abuse, 2014; Strang, 2015).

State laws vary in terms of who may administer naloxone. Some states shield professional first responders and nonprofessional Good Samaritans from criminal or civil liability if they administer naloxone or render comparable medical aid in the event of a drug overdose (Strang et al., 2006). Other states restrict administration of naloxone to licensed medical providers, trained law enforcement personnel, or other professional first responders.

Some Drug Court professionals may fear this practice could give the unintended message to participants that continued drug use is acceptable or anticipated. On the contrary, educating participants about drug overdose delivers a clear message about the potentially fatal consequences of continued drug abuse. Moreover, drug-abstinent participants may find themselves in the position of needing to save the life of a nonsobber family member or acquaintance. Preparing participants to respond effectively in such circumstances delivers the prosocial message that they have a responsibility to help their fellow citizens.

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VII. DRUG AND ALCOHOL TESTING

Drug and alcohol testing provides an accurate, timely, and comprehensive assessment of unauthorized³ substance use throughout participants' enrollment in the Drug Court.

- A. Frequent Testing
- B. Random Testing
- C. Duration of Testing
- D. Breadth of Testing
- E. Witnessed Collection
- F. Valid Specimens
- G. Accurate and Reliable Testing Procedures
- H. Rapid Results
- I. Participant Contract

A. Frequent Testing

Drug and alcohol testing is performed frequently enough to ensure substance use is detected quickly and reliably. Urine testing is performed at least twice per week until participants are in the last phase of the program and preparing for graduation. Tests that measure substance use over extended periods of time, such as ankle monitors, are applied for at least ninety consecutive days followed by urine or other intermittent testing methods. Tests that have short detection windows, such as breathalyzers or oral fluid tests, are administered when recent substance use is suspected or when substance use is more likely to occur, such as during weekends or holidays.

B. Random Testing

The schedule of drug and alcohol testing is random and unpredictable. The probability of being tested on weekends and holidays is the same as on other days. Participants are required to deliver a test specimen as soon as practicable after being notified that a test has been scheduled. Urine specimens are delivered no more than eight hours after being notified that a urine test has been scheduled. For tests with short detection windows, such as oral fluid tests, specimens are delivered no more than four hours after being notified that a test was scheduled.

C. Duration of Testing

Drug and alcohol testing continues uninterrupted to determine whether relapse occurs as other treatment and supervision services are adjusted.

³ Unauthorized substances include alcohol, illicit drugs, and addictive or intoxicating prescription medications that are taken without prior approval from the Drug Court and not during a medical emergency.

D. Breadth of Testing

Test specimens are examined for all unauthorized substances of abuse that are suspected to be used by Drug Court participants. Randomly selected specimens are tested periodically for a broader range of substances to detect new substances of abuse that might be emerging in the Drug Court population.

E. Witnessed Collection

Collection of test specimens is witnessed directly by a staff person who has been trained to prevent tampering and substitution of fraudulent specimens. Barring exigent circumstances, participants are not permitted to undergo independent drug or alcohol testing in lieu of being tested by trained personnel assigned to or authorized by the Drug Court.

F. Valid Specimens

Test specimens are examined routinely for evidence of dilution and adulteration.

G. Accurate and Reliable Testing Procedures

The Drug Court uses scientifically valid and reliable testing procedures and establishes a chain of custody for each specimen. If a participant denies substance use in response to a positive screening test, a portion of the same specimen is subjected to confirmatory analysis using an instrumented test, such as gas chromatography/mass spectrometry (GC/MS) or liquid chromatography/mass spectrometry (LC/MS). Barring staff expertise in toxicology, pharmacology, or a related discipline, drug or metabolite concentrations falling below industry- or manufacturer-recommended cutoff levels are not interpreted as evidence of new substance use or changes in participants' substance use patterns.

H. Rapid Results

Test results, including the results of confirmation testing, are available to the Drug Court within forty-eight hours of sample collection.

I. Participant Contract

Upon entering the Drug Court, participants receive a clear and comprehensive explanation of their rights and responsibilities related to drug and alcohol testing. This information is described in a participant contract or handbook and reviewed periodically with participants to ensure they remain cognizant of their obligations.

COMMENTARY

Certainty is one of the most influential factors for success in a behavior modification program (Harrell & Roman, 2001; Marlowe & Kirby, 1999). Outcomes improve significantly when detection of substance use is likely (Kilmer et al., 2012; Marques et al., 2014; Schuler et al., 2014), and participants receive incentives for abstinence and sanctions or treatment adjustments for positive test results (Hawken & Kleiman, 2009; Marlowe et al., 2005). Therefore, the success of any Drug Court will depend, in part, on the reliable monitoring of substance use. If a Drug

DRUG AND ALCOHOL TESTING

Court does not have accurate and timely information about whether participants are maintaining abstinence from alcohol and other drugs, the team has no way to apply incentives or sanctions correctly or to adjust treatment and supervision services accordingly. Drug and alcohol testing also serves other important therapeutic aims, such as helping to confirm clinicians' diagnostic impressions, providing objective feedback to participants about their progress or lack thereof in treatment, and assisting clinicians to challenge and resolve participant denial about the severity of their problems (American Society of Addiction Medicine (ASAM), 2010, 2013; DuPont & Selavka, 2008; DuPont et al., 2014; Srebnik et al., 2014).

Participants cannot be relied upon to self-disclose substance use accurately (Hunt et al., 2015). Studies consistently find that between 25% and 75% of participants in substance abuse treatment deny recent substance use when biological testing reveals a positive result (Auerbach, 2007; Harris et al., 2008; Hindin et al., 1994; Magura & Kang, 1997; Morral et al., 2000; Peters et al., 2015; Tassiopoulos et al., 2004). The accuracy of self-reporting is particularly low among individuals involved in the criminal justice system, presumably because they are likely to receive sanctions for substance use (Harrison, 1997; Peters et al., 2015). Although some clinicians may assume that the accuracy of self-report increases during the course of treatment, contrary evidence suggests participants may be less likely to acknowledge substance use after they have been enrolled in treatment for a period of time or have completed treatment (Wish et al., 1997). The longer participants are in treatment, the more staff come to expect and insist upon abstinence. For this reason, participants find it increasingly difficult to admit to substance abuse after they have been enrolled in treatment for several months (Davis et al., 2014; Nirenberg et al., 2013).

Best practices for conducting drug and alcohol testing vary considerably depending on whether a test is administered intermittently as opposed to continually, the length of the test's detection window, and the range of substances the test is capable of detecting. Some tests, such as urine or oral fluid tests, must be administered repeatedly, whereas others, such as sweat patches or ankle monitors, can measure substance use over extended periods of time. Most drug metabolites are detectable in urine for approximately two to four days, but are detectable in oral fluid for an average of twenty-four hours and in breath or blood for less than twelve hours (Auerbach, 2007; Cary, 2011; DuPont et al., 2014). Some tests, such as breathalyzers, can only assess for alcohol use, whereas urine tests can assess for a wide range of substances. These factors influence how the tests must be used to obtain useful results.

Urine testing is, by far, the most common methodology used in Drug Courts and probation programs. This is because urine is typically available in copious amounts, is relatively simple to collect, does not require elaborate sample preparation procedures, is inexpensive to analyze, and can be examined for many substances (Cary, 2011). Most studies, to date, have examined best practices for conducting urine testing with offenders; however, recent studies have begun to examine other testing methods in Drug Courts, including sweat patches and ankle monitors.

A. Frequent Testing

The more frequently Drug Courts and probation programs perform urine drug testing, the better their outcomes in terms of higher graduation rates and lower drug use and criminal recidivism (Banks & Gottfredson, 2003; Gottfredson et al., 2007; Griffith et al., 2000; Harrell et al., 1998; Hawken & Kleiman, 2009; Kinlock et al., 2013; National Institute on Drug Abuse, 2006). In focus groups, Drug Court participants consistently identified frequent drug and alcohol testing as being among the most influential factors for success in the program (Gallagher et al., 2015; Goldkamp et al., 2002; Saum et al., 2002; Turner et al., 1999; Wolfer, 2006).

The most effective Drug Courts perform urine drug testing at least twice per week for the first several months of the program (Carey et al., 2008). In a multisite study of approximately seventy Drug Courts, programs performing urine testing at least twice per week in the first phase produced 38% greater reductions in crime and were 61% more cost-effective than programs performing urine testing less frequently (Carey et al., 2012). Because the metabolites of most drugs of abuse are detectable in urine for approximately two to four days, testing less frequently leaves an unacceptable time gap during which participants can abuse substances and evade detection, thus leading to significantly poorer outcomes (Stitzer & Kellogg, 2008).

Recent studies have examined the impact of other testing methods in Drug Courts. The Secure Continuous Remote Alcohol Monitor (SCRAM) is an ankle device that can detect alcohol in sweat and transmits a wireless signal to a remote monitoring station. Preliminary evidence suggests the use of a SCRAM may

deter alcohol consumption and alcohol-impaired driving among recidivist driving-while-impaired (DWI) offenders if it is worn for at least ninety consecutive days (Flango & Cheesman, 2009; Tison et al., 2015). Another study found that adding sweat patches to urine testing did not improve outcomes in a Drug Court (Kleinpeter et al., 2010). However, that study did not examine the influence of sweat patches alone or as compared against urine testing. The study merely found that the addition of sweat patches did not improve outcomes beyond what was already being achieved from frequent urine drug testing.

Ethyl glucuronide (EtG) and ethyl sulfate (EtS) are metabolites of alcohol that can be detected in urine for longer periods of time than ethanol. The use of EtG or EtS can extend the time window for detecting alcohol consumption from several hours to several days (Cary, 2011). A recent randomized, controlled trial reported that participants completed the first two phases of a Drug Court significantly sooner when they were subjected to weekly EtG and EtS testing (Gibbs & Wakefield, 2014). The EtG and EtS testing enabled the Drug Court to respond more rapidly and reliably to instances of alcohol use, thus producing more efficient results. Importantly, EtG and EtS testing was determined in the same study to be superior to standard ethanol testing for detecting alcohol use occurring over weekends. Because some Drug Courts may not perform drug or alcohol testing on weekends, weekday tests capable of detecting weekend substance use are crucial.

As was noted previously, some drug or alcohol tests have short detection windows of twelve to twenty-four hours. This makes them generally unsuitable for use as the primary testing method in Drug Courts. Such tests can be used effectively, however, for spot-testing when recent use is suspected or during high-risk times, such as weekends or holidays. Evidence also suggests these tests can deter substance use effectively if they are administered on a daily basis. A statewide study in South Dakota found that daily breathalyzer testing significantly reduced failures to appear and rearrest rates among DWI offenders released on bail (Kilmer et al., 2012). In that study, daily breathalyzer testing appears to have been sufficient to deter alcohol consumption in the majority of cases without the need for additional services.

B. Random Testing

Drug and alcohol testing is most effective when performed on a random basis (ASAM, 2013; ASAM, 2010; Auerbach, 2007; Carver, 2004; Cary, 2011; Harrell & Kleiman, 2002; McIntire et al., 2007). If participants know in advance when they will be tested, they can adjust the timing of their usage or take other countermeasures, such as excessive fluid consumption, to defraud the tests (McIntire & Lessenger, 2007). Random drug testing elicits significantly higher percentages of positive tests than prescheduled testing, suggesting that many participants can evade detection if they have advance notice about when testing will occur (Harrison, 1997).

Random testing means the odds of being tested are the same on any given day of the week, including weekends and holidays. For example, if a participant is scheduled to be drug tested two times per week, then the odds of being tested should be two in seven (28%) on every day of the week. For this reason, Drug Courts should not schedule their testing regimens in seven-day or weekly blocks, which is a common practice. Assume, for example, that a participant is randomly selected for drug testing on Monday and Wednesday of a given week. If testing is scheduled in weekly blocks, then the odds of that same participant being selected again for testing on Thursday will be zero. In behavioral terms, this is referred to as a *respite* from detection, which can lead to increased drug or alcohol use owing to the absence of negative consequences (Marlowe & Wong, 2008).

The odds of being tested for drugs and alcohol should be the same on weekends and holidays as on any other day of the week (Marlowe, 2012). Weekends and holidays are high-risk times for drug and alcohol use (Kirby et al., 1995; Marlatt & Gordon, 1985). Providing a respite from detection during high-risk times reduces the randomness of testing and undermines the central aims of a drug-testing program (ASAM, 2013).

Limiting the time delay between notification of an impending drug or alcohol test and collection of the test specimen is essential (ASAM, 2013). If participants can delay provision of a specimen for even a day or two, they can rely on natural elimination processes to reduce drug and metabolite concentrations below cutoff levels. For participants who live in close proximity to the testing facility and do not have confirmed

scheduling conflicts, Drug Courts can reasonably expect samples to be delivered within a few hours of notification that a test has been scheduled (Cary, 2011). Barring exigent circumstances, participants should be required to deliver a urine specimen no more than eight hours after being notified that a urine test has been scheduled (Auerbach, 2007). This practice should give most participants ample time to meet their daily obligations and travel to the sample collection site, while also reducing the likelihood that metabolite concentrations will fall below cutoff levels. For tests with short detection windows of less than twenty-four hours, such as oral fluid tests, participants should be required to deliver a specimen no more than four hours after being notified that a test has been scheduled.

C. Duration of Testing

A basic tenet of behavior modification provides that the effects of any intervention should be assessed continually until all components of the intervention are completed (Rusch & Kazdin, 1981). This is the only way to know whether a participant is likely to relapse or regress after the program ends.

Drug Courts commonly decrease the intensity of treatment and supervision as participants make progress in the program. For example, the frequency of court hearings or case management sessions is commonly reduced as participants advance through successive phases. With a reduction of services comes the ever-present risk of relapse or other behavioral setback; therefore, drug and alcohol testing should continue uninterrupted to reveal any relapse as other components of the participants' treatment regimens are adjusted (Cary, 2011; Marlowe, 2011, 2012). Although research has not addressed the issue, logic dictates maintaining the frequency of drug and alcohol testing until participants are engaged in what will ultimately be their continuing-care or aftercare plan. This practice provides the greatest assurance that participants are likely to remain abstinent after program graduation.

D. Breadth of Testing

Drug Courts must test for the full range of substances that are likely to be used by participants in the program. Participants can easily evade detection of their substance use on many standard test panels—such as the National Institute on Drug Abuse five-panel test (NIDA-5) or a standard eight-panel test—simply by switching to other drugs of abuse that have similar psychoactive effects but are not detected by the test (ASAM, 2013). For example, heroin users can avoid detection by many standard test panels if they switch to pharmaceutical opioids, such as oxycodone or buprenorphine (Wish et al., 2012). Similarly, marijuana users can avoid detection by using synthetic cannabinoids, such as K2 or Spice, which were developed for the specific purpose of avoiding detection (Cary, 2014; Castaneto et al., 2014). Studies confirm that some marijuana users do switch to synthetic cannabinoids to evade detection by drug tests and then return to marijuana use after the testing regimen has been discontinued (Perrone et al., 2013). Because new substances of abuse are constantly being sought out by offenders to cheat drug tests, Drug Courts should select test specimens randomly and frequently and examine them for a wide range of potential drugs of abuse that might be emerging in their population (ASAM, 2013).

E. Witnessed Collection

Drug Court participants and probationers acknowledge engaging in widespread efforts to defraud drug and alcohol tests. These efforts include, but are not limited to, consuming excessive water to dilute the sample (dilution), adulterating the sample with chemicals intended to mask a positive result (adulteration), and substituting another person's urine or a look-alike sample that is not urine, such as apple juice (substitution) (Cary, 2011; McIntire & Lessenger, 2007). Collectively, these efforts are referred to as tampering. In focus groups, Drug Court participants reported being aware of several individuals in their program who tampered with drug tests on more than one occasion without being detected by staff (Goldkamp et al., 2002).

The most effective way to avoid tampering is to ensure that sample collection is witnessed directly by a trained and experienced staff person (ASAM, 2013; Cary, 2011). If substitution or adulteration is suspected, a new sample should be collected immediately under closely monitored conditions (McIntire et al., 2007). Staff members should be trained in how to implement countermeasures to avoid tampered test specimens. Examples of such countermeasures include searching participants' clothing for chemical adulterants or fraudulent samples, requiring participants to leave outerwear outside of the test-collection

room, and putting colored dye in the sink and toilet to prevent water from being used to dilute test specimens (McIntire & Lessenger, 2007).

If substitution or other efforts at tampering are suspected for a urine specimen, it may be useful to obtain an oral fluid specimen immediately as a secondary measure of substance use. Generally speaking, observing the collection of oral fluid closely is easier than for the collection of urine, and oral fluid tests are less susceptible to dilution than urine tests (Heltsley et al., 2012; Sample et al., 2010). However, because oral fluid testing has a shorter detection window than urine testing, a negative oral fluid test would not necessarily rule out recent drug use or the possibility of a tampered urine test.

Because specialized training is required to minimize tampering of test specimens, under most circumstances participants should be precluded from undergoing drug and alcohol testing by independent sources. In exigent circumstances, such as when participants live a long distance from the test collection site, the Drug Court might designate independent professionals or laboratories to perform drug and alcohol testing. As a condition of approval, these professionals should be required to complete formal training on the proper collection, handling, and analyses of drug and alcohol test samples among Drug Court participants or comparable criminal justice populations. Drug Courts are also required to follow generally accepted chain-of-custody procedures when handling test specimens (ASAM, 2013; Cary, 2011; Meyer, 2011). Therefore, if independent professionals or laboratories perform drug and alcohol testing, they must be trained carefully to follow proper chain-of-custody procedures.

F. Valid Specimens

Several low-cost analyses can be performed to detect adulterated or diluted test specimens (McIntire et al., 2007). The temperature of each urine specimen should be examined immediately upon collection to ensure it is consistent with an expected human body temperature. An unusual temperature might suggest the sample cooled down because it was collected at an earlier point in time, or was mixed with water that was too cold or too hot to be consistent with body temperature. Under normal conditions, urine specimens should be between 90⁰ and 100⁰ F within four minutes of collection, and a lower or higher temperature likely indicates a deliberate effort at deception (ASAM, 2013; Tsai et al., 1998).

Urine specimens should also be tested for creatinine and specific gravity. Creatinine is a metabolic product of muscle contraction that is excreted in urine at a relatively constant rate. A creatinine level below 20 mg/dL is rare and is a reliable indicator of an intentional effort at dilution or excessive fluid consumption barring unusual medical or metabolic conditions (ASAM, 2013; Cary, 2011; Jones & Karlsson, 2005; Katz et al., 2007). Specific gravity reflects the amount of solid substances that are dissolved in urine. The greater the specific gravity, the more concentrated the urine; and the lower the specific gravity, the closer its consistency to water. The normal range of specific gravity for urine is 1.003 to 1.030, and a specific gravity of 1.000 is essentially water. Some experts believe a specific gravity below 1.003 reflects a diluted sample (Katz et al., 2007). Although this analysis, by itself, may not be sufficient to prove excessive fluid consumption, dilution is likely to have occurred if the specific gravity is low and accompanies other evidence of tampering or invalidity, such as a low creatinine level or temperature. Several commercially available test strips, such as Adulcheck and Intect, have also been shown to reliably detect dilution or adulteration of urine test samples (Dasgupta et al., 2004; Mikkelsen & Ash, 1988).

G. Accurate and Reliable Testing Procedures

To be admissible as evidence in a legal proceeding, drug and alcohol test results must be derived from scientifically valid and reliable methods (Meyer, 2011). Appellate courts have recognized the scientific validity of several commonly used methods for analyzing urine, including gas chromatography/mass spectrometry (GC/MS), liquid chromatography/tandem mass spectrometry (LC/MS/MS), the enzyme multiple immunoassay technique (EMIT), and some sweat, oral fluid, hair, and ankle-monitor tests (Meyer, 2011).

Tests such as GC/MS and LC/MS/MS are referred to as instrumented tests, laboratory-based tests, or confirmation tests. These tests have a higher degree of scientific precision than immunoassay tests, point of collection tests (POCT), or screening tests, such as on-site test cups or instant test strips. If a participant

denies substance use in the face of a positive screening test, courts will typically require, and toxicology experts recommend, performing confirmation testing using GC/MS or a similar instrumented technique (ASAM, 2013; Cary, 2011). Confirmation with an instrumented test virtually eliminates the odds of a false-positive result, assuming the sample was collected and stored properly (Auerbach, 2007; Peat, 1988). Drug Courts commonly require participants to pay the cost of confirmation tests if the initial screening result is confirmed (Cary, 2011; Meyer, 2011). Confirmation testing should be performed on a portion of the original test specimen. If confirmation testing is performed on a different specimen that was collected at a later point in time, a conflicting result might not reflect a failure to confirm but rather differences in the detection windows for the tests or the metabolic processes of the participant.

Drug Courts must follow generally accepted chain-of-custody procedures when handling test specimens (ASAM, 2013; Cary, 2011; Meyer, 2011). They need to establish a reliable paper trail identifying each professional who handled the specimen from collection through laboratory analysis to reporting of the results. Establishing a proper chain of custody requires sufficient labeling and security measures to provide confidence the specimen belongs to the individual identified on the record and the specimen was transported and stored according to generally accepted laboratory procedures and manufacturer recommendations.

Some Drug Courts interpret changes in quantitative levels of drug metabolites as evidence that new substance use has occurred or a participant's substance use pattern has changed. Unless a Drug Court has access to an expert trained in toxicology, pharmacology, or a related discipline, such practices should be avoided. Quantitative metabolite levels can vary considerably based on a number of factors, including the total fluid content in urine or blood (Cary, 2004; Schwilke et al., 2010). Moderate changes in participants' fluid intake or fluid retention could lead Drug Courts to miscalculate substance use patterns. Most drug and alcohol tests used in Drug Courts were designed to be *qualitative*, meaning they were designed to determine whether a drug or drug metabolite is present at levels above a prespecified concentration level. The cutoff concentration level is calculated empirically to maximize the true-positive rate, true-negative rate, or classification rate. When Drug Courts engage in quantitative analyses, they are effectively altering the cut-off score and making the results less accurate.

Some Drug Courts have difficulty interpreting positive cannabinoid (marijuana) test results. Because cannabinoids are lipid-soluble (i.e., bind to fat molecules), they may be excreted more slowly than other substances of abuse. This has caused confusion about when a positive cannabinoid result may be interpreted as evidence of new use as opposed to residual use from an earlier episode. A participant is highly unlikely to produce a cannabinoid-positive urine result above 50 ng/mL after more than ten days following cessation of chronic usage or for more than three to four days following a single-use event (Cary, 2005). Therefore, a Drug Court would be justified in considering the first two weeks of enrollment to be a grace period during which there would be no sanctions for positive cannabinoid test results. However, subsequent positive tests may be interpreted as evidence of new cannabis use and dealt with accordingly. Moreover, once a participant has produced two consecutive cannabinoid-negative urine specimens (called an *abstinence baseline*), a subsequent cannabinoid-positive test may be interpreted as new use (Cary, 2005). Some Drug Courts or laboratories may employ a lower cutoff level of 20 ng/mL for cannabis metabolites. Using this lower cutoff, thirty days is sufficient to establish a presumptive abstinence baseline even for chronic users (Cary, 2005); in the majority of cases, twenty-one days should be sufficient.

Some participants may attempt to attribute a positive cannabinoid test to passive inhalation or second-hand smoke. This excuse should not be credited. The likelihood of passive inhalation triggering a positive cannabinoid test is negligible (Cone et al., 2014; Law et al., 1984; Katz et al., 2007; Niedbala et al., 2005). Moreover, because Drug Court participants are usually prohibited from associating with people who are engaged in substance use, passive inhalation may be viewed as a violation of this central prohibition, thus meriting an additional sanction (Marlowe, 2011).

H. Rapid Results

In addition to certainty, timing is one of the most influential factors for success in a behavior modification program (Harrell & Roman, 2001; Marlowe & Kirby, 1999). The sooner sanctions are delivered after an infraction and incentives delivered after an achievement, the better the results. Because sanctions and

incentives are imposed routinely on the basis of drug and alcohol test results, the Drug Court team needs test results before participants appear for status hearings.

A study of approximately seventy Drug Courts reported significantly greater reductions in criminal recidivism and significantly greater cost benefits when the teams received drug and alcohol test results within forty-eight hours of sample collection (Carey et al., 2012). Drug Courts that received test results within forty-eight hours were 73% more effective at reducing crime and 68% more cost-effective than Drug Courts receiving test results after longer delays. Ordinarily, negative test results should take no longer than one business day to produce, and positive results should require no more than two days if confirmation testing is requested (Cary, 2011; Robinson & Jones, 2000).

I. Participant Contract

Outcomes are significantly better when Drug Courts specify their policies and procedures clearly in a participant manual or handbook (Carey et al., 2012). Criminal defendants are significantly more likely to react favorably to an adverse judgment if they were given advance notice about how such judgments would be made (Burke & Leben, 2007; Frazer, 2006; Tyler, 2007). Drug Courts can enhance participants' perceptions of fairness substantially and reduce avoidable delays from contested drug and alcohol tests by describing their testing procedures and requirements in a participant contract or handbook.

Below are examples of provisions that should be included in a participant contract to address many of the best practices discussed above. For participants with limited educational histories, the language may need to be simplified and the requirements explained orally. Repeat the information periodically to ensure participants understand their rights and obligations.

- Drug and alcohol testing will be performed frequently and on a random basis throughout your enrollment in the Drug Court.
- Drug and alcohol testing will be performed on weekends and holidays.
- Drug and alcohol testing will be performed by a laboratory or program approved by the Drug Court.
- Because cannabinoids (a byproduct of marijuana) may persist in the body for several days, marijuana users have a two-week grace period following enrollment during which no sanctions will be given for positive cannabinoid test results. However, after two weeks positive cannabinoid tests will be presumed to reflect new marijuana use. Participants bear the burden of establishing a convincing alternative explanation for such results. After you have had two consecutive cannabinoid-negative urine specimens, the Drug Court will presume that subsequent positive cannabinoid results reflect new use.
- You must arrive at the testing facility as soon as possible after being notified that a test has been scheduled. You will be sanctioned for an unexcused failure to arrive within eight hours of being notified that a urine test has been scheduled or within four hours for tests that have short detection windows, such as breath or oral fluid tests.
- A staff person will directly observe the collection of test specimens. The staff person will be the same gender as you unless you, your defense attorney or your therapist request otherwise.
- Failure to provide a test specimen or providing an insufficient volume of fluid for analysis is an infraction of the rules of the program and will be sanctioned accordingly. You will be given a sufficient time (up to one hour) to deliver a urine specimen and allowed to drink up to one cup of water in the presence of staff.
- You may not drink any fluid excessively before testing and must avoid environmental contaminants, over-the-counter medications, or foods that can reduce the accuracy of the tests. Potential contaminants that you need to avoid are [provide list of contaminants].
- You may be subjected to immediate spot testing if the Drug Court has reason to suspect recent use or during high-risk times such as weekends or holidays.

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- You have the right to challenge the results of a screening test and to request proof that an adequate chain of custody was established for your specimen. The Drug Court will rely on the results of an instrumented or laboratory-based test in confirming whether substance use has occurred. You may be charged the cost of the confirmation test if a screening test is confirmed.
- You will be sanctioned for providing diluted, adulterated, or substituted test specimens. Urine specimens below 90° F, above 100° F, or that have a creatinine level below 20 mg/dL will be presumed to be diluted or fraudulent. Participants bear the burden of establishing a convincing alternative explanation for such results. Under such circumstances, you may receive two sanctions, one for the substance use and one for the effort at deception.
- You will be sanctioned for using synthetic substances such as K2 or Spice that are designed to avoid detection by standard drug tests. Switching to a new substance of abuse (for example, switching from heroin to an unauthorized prescription opioid) will be presumed to be an effort to defraud the drug test. You may receive two sanctions in such circumstances, one for the substance use and one for the effort at deception.
- You will be sanctioned for associating with other people who are engaged in substance use or for exposing yourself to passive inhalation or secondhand smoke.

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VIII. MULTIDISCIPLINARY TEAM

A dedicated multidisciplinary team of professionals manages the day-to-day operations of the Drug Court, including reviewing participant progress during pre-court staff meetings and status hearings, contributing observations and recommendations within team members' respective areas of expertise, and delivering or overseeing the delivery of legal, treatment and supervision services.

- A. Team Composition**
- B. Pre-Court Staff Meetings**
- C. Sharing Information**
- D. Team Communication and Decision Making**
- E. Status Hearings**
- F. Team Training**

A. Team Composition

The Drug Court team comprises representatives from all partner agencies involved in the creation of the program, including but not limited to a judge or judicial officer, program coordinator, prosecutor, defense counsel representative, treatment representative, community supervision officer, and law enforcement officer.

B. Pre-Court Staff Meetings

Team members consistently attend pre-court staff meetings to review participant progress, determine appropriate actions to improve outcomes, and prepare for status hearings in court. Pre-court staff meetings are presumptively closed to participants and the public unless the court has a good reason for a participant to attend discussions related to that participant's case.

C. Sharing Information

Team members share information as necessary to appraise participants' progress in treatment and compliance with the conditions of the Drug Court. Partner agencies execute memoranda of understanding (MOUs) specifying what information will be shared among team members. Participants provide voluntary and informed consent permitting team members to share specified data elements relating to participants' progress in treatment and compliance with program requirements. Defense attorneys make it clear to participants and other team members whether they will share communications from participants with the Drug Court team.

D. Team Communication and Decision Making

Team members contribute relevant insights, observations, and recommendations based on their professional knowledge, training, and experience. The judge considers the perspectives of all team members before making decisions that affect participants'

welfare or liberty interests and explains the rationale for such decisions to team members and participants [see Standard III, Roles and Responsibilities of the Judge].

E. Status Hearings

Team members attend status hearings on a consistent basis. During the status hearings, team members contribute relevant information or recommendations when requested by the judge or as necessary to improve outcomes or protect participants' legal interests.

F. Team Training

Before starting a Drug Court, team members attend a formal preimplementation training to learn from expert faculty about best practices in Drug Courts and develop fair and effective policies and procedures for the program. Subsequently, team members attend continuing education workshops on at least an annual basis to gain up-to-date knowledge about best practices on topics including substance abuse and mental health treatment, complementary treatment and social services, behavior modification, community supervision, drug and alcohol testing, team decision making, and constitutional and legal issues in Drug Courts. New staff hires receive a formal orientation training on the Drug Court model and best practices in Drug Courts as soon as practicable after assuming their position and attend annual continuing education workshops thereafter.

COMMENTARY

The Drug Court team is a multidisciplinary group of professionals responsible for administering the day-to-day operations of a Drug Court, including reviewing participant progress during pre-court staff meetings and status hearings, contributing observations and recommendations within team members' respective areas of expertise, and delivering or overseeing the delivery of legal, treatment, and supervision services (Hardin & Fox, 2011). Some Drug Courts may have additional governing bodies such as Steering Committees that are not involved in the daily operations of the program, but provide oversight on policies and procedures, negotiate MOUs between partner agencies, garner political and community support for the Drug Court, or engage in fundraising. Researchers have examined the influence of the multidisciplinary Drug Court team on participant outcomes but have not addressed the influence of other governing bodies.

A. Team Composition

Studies reveal the composition of the Drug Court team has a substantial influence on outcomes. Drug Courts produce significantly greater reductions in criminal recidivism and are significantly more cost-effective when the following professionals are dedicated members of the Drug Court team and participate regularly in pre-court staff meetings and status hearings (Carey et al., 2008, 2012; Cissner et al., 2013; Rossman et al., 2011; Shaffer, 2010):

- *Judge*—Typically a trial court judge leads the Drug Court team; however, in some jurisdictions a nonjudicial officer such as a magistrate or commissioner may preside over the Drug Court. Nonjudicial officers usually report directly to a judge and require judicial authorization for actions that affect participants' liberty interests such as jail sanctions or discharge from the program. No study has compared outcomes between judges and nonjudicial officers.
- *Program Coordinator*—Typically a court administrator or clerk serves as the coordinator for the Drug Court program; however, some Drug Courts may employ a senior probation officer, case manager, or clinician as the coordinator. Among many other duties, the coordinator is responsible for maintaining

accurate and timely records and documentation for the program, overseeing fiscal and contractual obligations, facilitating communication between team members and partner agencies, ensuring policies and procedures are followed, overseeing collection of performance and outcome data, scheduling court sessions and staff meetings, and orienting new hires.

- *Prosecutor*—Typically an assistant district attorney serves on the team. Among other duties, the prosecutor advocates on behalf of public safety, victim interests, and holding participants accountable for meeting their obligations in the program. The prosecutor may also help to resolve other pending legal cases that impact participants' legal status or eligibility for Drug Court.
- *Defense Attorney*—Typically an assistant public defender or private defense attorney specializing in Drug Court cases serves on the team. Among other duties, the defense attorney ensures participants' constitutional rights are protected and advocates for participants' stated legal interests. Defendants are usually represented by a public defender or private defense attorney in proceedings leading up to their entry into Drug Court. After entry, participants may retain their previous defense counsel, provide informed consent to be represented by a defense representative serving on the Drug Court team, or consent to be represented jointly by private defense counsel and the defense representative. In cases of joint representation, the defense representative typically handles most day-to-day issues relating to Drug Court participation, but private counsel may step in if the participant faces a potential jail sanction or discharge from the program (Freeman-Wilson et al., 2003; Tobin, 2012).

In postconviction Drug Courts, participation in the program is a condition of probation or part of a criminal sentence. Ordinarily, participants are not entitled to defense representation at the postconviction stage unless they face a potential jail sanction or revocation of probation (Meyer, 2011a). Nevertheless, postconviction Drug Courts should include a defense representative on their team because studies indicate defense involvement improves outcomes significantly (Carey et al., 2012; Cissner et al., 2013; National Association of Drug Court Professionals [NADCP], 2009). Evidence suggests participants may be more likely to perceive Drug Court procedures as fair when a dedicated defense attorney represents their interests in team meetings and status hearings (Frazer, 2006), and greater perceptions of fairness are consistently associated with better outcomes in Drug Courts and other problem-solving courts (Berman & Gold, 2012; Burke, 2010; Gottfredson et al., 2007; Rossman et al., 2011).

Some Drug Courts require participants to waive defense representation as a condition of entry. Although no case has addressed this issue squarely in the context of Drug Court, the weight of legal authority suggests defendants and probationers are entitled to withdraw such waivers and reassert their right to counsel at critical stages in the proceedings such as when they face a potential jail sanction or probation revocation (McKaskle v. Wiggins, 1984; Menefield v. Borg, 1989; Robinson v. Ignacio, 2004; State v. Pitts, 2014). Regardless of the legality of such waivers, defense representation should be encouraged rather than discouraged in Drug Courts because doing so is associated with significantly better outcomes and ensures participants' due process rights are protected (Hora & Stalcup, 2008; NADCP, 2009).

- *Community Supervision Officer*—Typically a probation officer or pretrial services officer serves on the team; however, some Drug Courts may rely on law enforcement or specially trained case managers or social service professionals to provide community supervision. Duties of the community supervision officer may include performing drug and alcohol testing, conducting home or employment visits, enforcing curfews and travel restrictions, and delivering cognitive-behavioral interventions designed to improve participants' problem-solving skills and alter dysfunctional criminal-thinking patterns (Harberts, 2011).
- *Treatment Representative*—Typically an addiction counselor, social worker, psychologist, or clinical case manager serves on the team. In many Drug Courts, participants can be referred to multiple treatment agencies or providers for substance abuse treatment and other complementary services such as mental health counseling or vocational rehabilitation. Because it is unwieldy to have multiple providers attend pre-court staff meetings and status hearings, many Drug Courts will designate one or two treatment professionals to serve as treatment representatives on the Drug Court team (Carey et al., 2012). The treatment representatives receive clinical information from programs treating Drug Court

participants, report that information to the Drug Court team, and contribute clinical knowledge and expertise during team deliberations.

- *Law Enforcement Officer*—Typically a police officer, deputy sheriff, highway patrol officer, or jail official serves on the team. Law enforcement is often the eyes and ears of Drug Court on the street, observing participant behavior and interacting with participants in the community. Law enforcement may also assist with home and employment visits, and serves as a liaison between the Drug Court and the police department, sheriff's office, jail, and correctional system.

Drug Courts may include other community representatives on their team as well, such as peer mentors, vocational advisors, or sponsors from the self-help recovery community. Studies have not examined the impact of including such persons on the Drug Court team; however, anecdotal reports suggest this practice can enhance team decision making and effectiveness (Taylor, 2014). As a condition of federal grant funding and funding from many states, Drug Courts may also be required to include an evaluator on their team beginning in the planning stages for the program and continuing during implementation. This practice helps to ensure Drug Courts collect reliable performance data to report to grant-making authorities and is generally advisable for all Drug Courts to ensure good-quality program monitoring and evaluation [see Standard X, Monitoring and Evaluation]. Finally, Drug Courts may be advised to include a nurse or physician on their team if they treat substantial numbers of participants requiring medication-assisted treatment or suffering from co-occurring medical or mental health disorders.

B. Pre-Court Staff Meetings

The Drug Court model requires Drug Courts to hold pre-court staff meetings—commonly referred to as *staffings* or *case reviews*—to review participant progress, develop a plan to improve outcomes, and prepare for status hearings in court (Hardin & Fox, 2011; NADCP, 1997; Roper & Lessenger, 2007). Not every participant is discussed in every meeting; however, staffings are held frequently enough (typically weekly or at the same frequency as status hearings) to ensure the team has an opportunity to consider the needs of each case.

Consistent attendance by all team members at staffings is associated with significantly better outcomes (Carey et al., 2012; Cissner et al., 2013; Rossman et al., 2011; Shaffer, 2010). A multisite study of approximately seventy Drug Courts found that programs were 50% more effective at reducing recidivism when all team members—the judge, prosecutor, defense representative, program coordinator, treatment representative, law-enforcement representative, and community supervision officer—attended staffings on a consistent basis (Carey et al., 2008, 2012). Drug Courts were nearly twice as cost-effective when defense counsel attended staffings consistently, and were more than twice as effective at reducing recidivism when the program coordinator, treatment representative, and law enforcement representative attended staffings consistently (Carey et al., 2012).

In most Drug Courts, staffings are presumptively closed. Discussions are not transcribed or recorded and the meeting is not open to the public or to participants unless the court has a good reason to allow a participant to attend discussions related to his or her case. Few appellate opinions have addressed the constitutionality or legality of closing staffings. In a recent opinion, the Washington State Supreme Court—which traditionally holds a very dim view of off-the-record proceedings—ruled that staffings may be presumptively closed at the discretion of the Drug Court judge (*State of Washington v. Sykes*, 2014). The Court analogized staffings to *pre-court conferences* in which attorneys commonly meet with the judge in chambers to clarify what legal issues are under contention, determine which facts are in dispute, and address other practical or collateral matters necessary to achieve a fair and efficient resolution of the case, such as scheduling witnesses or issuing discovery orders. In line with this reasoning, staffings may be closed so long as no final decisions are reached concerning disputed facts or legal issues in the case, and the judge recites in open court what decisions, if any, were made during the staffing. A closed staffing may not result in a binding order or factual conclusion related to a contested matter (Meyer, 2011a). Contested matters must be addressed and resolved in open court during status hearings or related due process hearings such as termination hearings or probation violation hearings.

Studies have not determined whether closed staffings produce more favorable outcomes than open staffings. The rationale for closing staffings derives largely from empirical studies and ethical analyses conducted in the context of psychotherapy progress notes and case conferences. For example, the Health Insurance Portability and Accountability Act (HIPAA) of 1996 grants broad access for patients to their health records, yet provides a lone exception for psychotherapy progress notes (45 C.F.R. §§ 164.508(a)(2) & 164.524; U.S. Dept. of Health & Human Services [U.S. DHHS], 2003; Wooten v. Duane Reade, 2009). Psychotherapy notes receive heightened protection against patient access, in part, because they often contain sensitive information provided by collateral sources, such as family members and friends (U.S. DHHS, 2003). If participants could gain access to this information, collateral sources might not be forthright in providing sensitive information about matters which are critical for delivering effective treatment, such as providing accurate histories of participants' substance abuse patterns, criminality, or related conduct (Stasiewicz et al., 2008). Studies have also reported that patients can be harmed psychologically by receiving unfettered access to their therapists' diagnostic impressions and conclusions (Lajeunesse & Lussier, 2010; Ross & Lin, 2003; Sergeant, 1986; Short, 1986; Westin, 1977). Sensitive clinical information must be communicated to patients in a cautious, empathic, and understandable manner to avoid causing psychological distress, embarrassment, confusion, or other untoward reactions (McFarlane et al., 1980; Miller et al., 1987).

Participant attendance at staffings might also inhibit free flow of information among staff, which is necessary to achieve productive aims. Treatment representatives, for example, may be reluctant to discuss their concerns about a participant's prognosis in front of the participant. Probation officers might similarly be reticent to recommend sanctions for participants in response to infractions. It is one thing for sanctions to be imposed by the team as a whole, but quite another for an individual staff member to be identified as the person who first proposed the sanction. Closed staffings allow team members to freely consider alternative courses of action that may or may not be adopted ultimately by the team.

Although staffings are presumptively closed, the judge and team may conclude they have a good reason for a participant to attend discussions related to that participant's case. For example, the team might wish to discuss highly sensitive matters with a participant in private, such as a history of childhood sexual abuse or positive HIV test result. Drug Courts are encouraged to include participants in staffings when clinically indicated or necessary to protect a participant from serious harm resulting from public disclosure of highly sensitive treatment information.

C. Sharing Information

Participants and staff rate communication among team members as one of the most important factors for success in Drug Courts (Frazer, 2006; Gallagher et al., 2015; Lloyd et al., 2014). Participants complain frequently that they are forced to repeat the same information to different professionals and to comply with excessive and inconsistent mandates stemming from different agencies (Goldkamp et al., 2002; Saum et al., 2002; Turner et al., 1999). Ongoing communication among staff ensures participants receive consistent messages, reduces unwarranted burdens on participants, and prevents participants from falling through the cracks or eluding responsibility for their actions by providing different information selectively to different team members.

Contrary to some misconceptions, the HIPAA and other applicable confidentiality statutes (e.g., Confidentiality of Substance Abuse Patient Records, 42 C.F.R. Part 2) do *not* prohibit treatment professionals or criminal justice professionals from sharing information related to substance abuse and mental health treatment (Matz, 2014; Meyer, 2011b). Rather, these statutes control how and under what circumstances such information may be disclosed (U.S. DHHS, 2003). Treatment professionals are generally permitted to share confidential treatment information with criminal justice professionals pursuant to a voluntary, informed, and competent waiver of a patient's confidentiality and privacy rights (45 C.F.R. §164.502(a)) or pursuant to a court order (45 C.F.R. §164.512(e)).

The scope of the disclosure must be limited to the minimum information necessary to achieve the intended aims of the disclosure (45 C.F.R. §§164.502(b) & 164.514(d)). In Drug Courts, team members may ordinarily share information pursuant to a valid waiver to the degree necessary to ensure that participants are progressing adequately in treatment and complying with other conditions of the program (Meyer,

2011b). At a minimum, the following data elements are required by all Drug Court team members to appraise participant progress and compliance or noncompliance with the conditions of Drug Court:

- Assessment results pertaining to a participant's eligibility for Drug Court and treatment and supervision needs
- Attendance at scheduled appointments
- Drug and alcohol test results, including efforts to defraud or invalidate said tests
- Attainment of treatment plan goals, such as completion of a required counseling regimen
- Evidence of symptom resolution, such as reductions in drug cravings or withdrawal symptoms
- Evidence of treatment-related attitudinal improvements, such as increased insight or motivation for change
- Attainment of Drug Court phase requirements, such as obtaining and maintaining employment or enrolling in an educational program
- Compliance with electronic monitoring, home curfews, travel limitations, and geographic or association restrictions
- Adherence to legally prescribed and authorized medically assisted treatments
- Procurement of unauthorized prescriptions for addictive or intoxicating medications
- Commission of or arrests for new offenses
- Menacing, threatening, or disruptive behavior directed at staff members, participants or other persons

To be legally valid, an informed consent document must specify what data elements may be shared, with whom, and for what authorized period of time (Meyer, 2011b). Therefore, the above data elements and any other information that may be shared among team members should be listed in releases of information or confidentiality waivers executed by Drug Court participants (Meyer, 2011b). If the scope of the disclosure is not enumerated clearly, then the waiver may not be knowing or informed—and thus may be legally invalid. Consent documents must also indicate which professionals are authorized to receive the information, what steps participants must take to revoke consent, and when the consent expires. Expiration of consent may be predicated upon a specific event, such as discharge from Drug Court, as opposed to a specific date or time frame (Meyer, 2011b). Finally, recipients of confidential information must be put on notice that they are only permitted to redisclose information to additional parties under carefully specified and approved conditions. MOUs between partner agencies—referred to as business associate contracts pursuant to HIPAA—must state clearly that confidential information may not be redisclosed to additional parties outside of the Drug Court without the express written permission of the participant and may not be used to prosecute new charges against the participant.

Assuming a participant has executed a valid waiver of his or her privacy and confidentiality rights, Drug Court team members are permitted, and indeed may be required, to share covered information in the course of performing their professional duties. Confidentiality and privacy rights belong to the participant, not to staff, and may be waived freely and voluntarily in exchange for receiving anticipated benefits, such as gaining access to effective treatment or avoiding a criminal record or jail sentence (Melton et al., 2007). Failing to abide by a valid confidentiality waiver could, under some circumstances, be a breach of a staff person's professional responsibilities to the participant.

Staff persons also have ethical obligations to other Drug Court team members. If a staff person knowingly withholds relevant information about a participant from other team members, this omission could inadvertently interfere with the participant's treatment goals, endanger public safety, or undermine the functioning of the Drug Court team. All agencies involved in the administration of a Drug Court should, therefore, execute MOUs specifying what data elements will be shared among team members (Harden & Fox, 2011). The data elements listed above might be included in such MOUs to clarify the obligations of each professional on the team.

If a staff person questions the validity or legality of a consent waiver, that staff person should raise this concern with the Drug Court team and make it clear that he or she may withhold relevant progress information until the matter is resolved. This course of action puts the Drug Court team on notice that important information may not be forthcoming and reduces the likelihood that mistaken actions will be taken based on erroneous or incomplete information.

Controversy surrounds the question of whether defense representatives should report infractions by participants to the Drug Court team. In most instances, infractions come to the attention of the team from sources other than defense counsel, such as positive drug tests or progress reports from treatment providers or probation officers. In some instances, however, participants may self-disclose infractions to defense representatives which would otherwise go undetected by the program.

Some defense experts advise against disclosing such communications because doing so may violate the attorney's ethical duty to advocate for the participant's stated legal interests, which are to be distinguished from the participant's *best* interests (Boldt, 1998; National Association of Criminal Defense Lawyers [NACDL], 2009). Other defense experts take the contrary position that withholding such information may undermine the defense representative's trustworthiness and credibility with the team. If team members know or suspect that defense counsel is shielding important information from them, they may discount recommendations from that defense expert as one-sided or nonobjective or may withhold information of their own (Tobin, 2012). In the absence of empirical evidence or legal precedent to guide the decision, defense representatives should make clear their position and the rationale for that position to participants and team members from the outset of each case (Freeman-Wilson et al., 2003). Participants have a right to know whether some confidences shared with defense representatives may be disclosed to other staff members, and team members have a right to know whether some information may not be available to them for decision making.

D. Team Communication and Decision Making

Before the advent of Drug Courts, studies of *courtroom workgroups* raised concerns about relying on multidisciplinary teams to manage criminal and civil cases. In response to overwhelming court dockets in the 1980s, some jurisdictions appointed teams of professionals—commonly including a judge, defense attorney, prosecutor, court clerk, probation officer, and bailiff—to process certain types of cases more efficiently, such as drug possession cases and child maltreatment cases. Observational studies revealed these workgroups tended to routinize their procedures to speed case processing, often at the expense of applying evidence-based practices or adapting dispositions to the needs and risk levels of litigants (Haynes et al., 2010; Knepper & Barton, 1997; Lipetz, 1980). Teaming up as a group did not necessarily improve outcomes and in some cases may have undermined litigants' due process rights. Drug Courts must not, in the interest of expediency, allow assembly-line procedures or groupthink mindsets to interfere with their adherence to due process and best practices.

Drug Courts are properly characterized as nonadversarial programs, meaning participants waive some, but not all, adversarial trial rights as a condition of entry, including the right to a speedy trial and to refuse to provide self-incriminating information (Hora & Stalcup, 2008; NADCP, 1997). Moreover, unlike traditional adversarial proceedings, the Drug Court judge speaks directly to participants rather than through legal counsel and takes an active role in supervising cases. The term nonadversarial does *not*, however, imply that team members relinquish their professional roles or responsibilities (Holland, 2010; Hora & Stalcup, 2008). Prosecutors continue to advocate on behalf of public safety, victim interests, and participant accountability; defense counsel continue to advocate for participants' legal rights; and treatment providers continue to advocate for effective and humane treatment (Freeman-Wilson et al., 2003; Holland, 2010; Tobin, 2012). In other words, the term *nonadversarial* does not have the same meaning as *nonadvocacy*. The principal distinction in Drug Courts is that advocacy occurs primarily in staffings as opposed to court hearings, reserving the greater share of court time for intervening with participants rather than arbitrating uncontested facts or legal issues (Christie, 2014; Portillo et al., 2013).

How Drug Court teams make decisions in this nonadversarial climate has constitutional implications. Due process and judicial ethics require Drug Court judges to exercise independent discretion when resolving factual controversies, ordering conditions of treatment and supervision, and administering sanctions and

incentives that affect participants' liberty interests (Hora & Stalcup, 2008; Meyer, 2011c; Meyer & Tauber, 2011). The judge may not delegate these decisions to the Drug Court team or acquiesce to majority rule [see Standard III, Roles and Responsibilities of the Judge]. The judge must, however, consider arguments from all sides of a controversy (typically from the defense and prosecution) before rendering a decision and must hear evidence from scientific experts if the subject matter of the controversy is beyond the common knowledge of laypersons (Hora & Stalcup, 2008; Meyer, 2011a). Information relating to addiction science and substance abuse treatment is typically beyond the knowledge of laypersons; therefore, this information must usually be introduced or explained by a qualified expert (e.g., Federal Rule of Evidence 702, 2015).

In Drug Courts, the multidisciplinary team serves essentially as a panel of "expert witnesses" providing legal and scientific expertise for the judge (Bean, 2002; Hora & Stalcup, 2008). Team members have an obligation to contribute relevant observations and insights and to offer suitable recommendations based on their professional knowledge, experience, and training. A team member who remains silent in staffings or defers habitually to group consensus is violating his or her professional obligations to participants and to the administration of justice (Freeman-Wilson et al., 2003; Holland, 2010; NACDL, 2009; Tobin, 2012). The judge may ultimately overrule a team member's assertions, but this fact does not absolve the team member from articulating and justifying an informed opinion.

Studies have identified effective communication strategies that can enhance team decision making in Drug Courts. For example, researchers have improved team decision-making skills in several Drug Courts using the NIATx (Network for the Improvement of Addiction Treatment) Organizational Improvement Model (Melnick et al., 2014a, 2014b; Wexler et al., 2012). The NIATx model seeks to create a climate of psychological safety by teaching team members to articulate divergent views in a manner that is likely to be heeded by fellow team members. Examples of NIATx techniques include the following (Melnick et al., 2014b):

- *Avoid Ego-Centered Communications*—Focus statements on the substantive issue at hand rather than attempting to be "right" or win an argument.
- *Avoid Downward Communication*—Ensure that all team members, regardless of status or authority, have an equal opportunity to speak.
- *Practice Attentive Listening*—Hear all aspects of a team member's statements before thinking about or forming a response.
- *Reinforce Others' Statements*—Express appreciation for a team member's input before making counterarguments or changing the subject.
- *Find Common Ground*—Acknowledge areas of agreement among team members before making counterarguments.
- *Reframe Statements Neutrally*—Restate a position in a manner that minimizes counterproductive affect such as anger or frustration.
- *Ensure Inclusiveness*—Ensure that all team members weigh in on subjects within their area of expertise or experience.
- *Show Understanding*—Restate others' positions to demonstrate accurate understanding.
- *Engage in Empathic Listening*—Imagine oneself in other team members' positions to understand issues from their perspective.
- *Sum Up*—The judge should recap the various arguments and positions, assure the team that all positions were considered carefully, and explain his or her rationale for reaching a conclusion or tabling the matter pending further information.

Preliminary studies in more than ten Drug Courts found that training Drug Court teams on the NIATx model enhanced team communication skills (Melnick et al., 2014b), increased staff job satisfaction (Melnick et al., 2014a), and improved program efficiency, leading to higher admission rates, shorter wait times for treatment, and reduced no-show rates at scheduled appointments (Wexler et al., 2012).

E. Status Hearings

Status hearings are critical components of Drug Courts (NADCP, 1997). In status hearings, participants interact with all team members in the same proceeding, the judge speaks personally with each participant, and incentives, sanctions and treatment adjustments are administered in accordance with participants' progress or lack thereof in treatment (Roper & Lessenger, 2007). A substantial body of research establishes convincingly that better outcomes are achieved when status hearings are held biweekly (every two weeks) or more frequently at least during the first phase of Drug Court (Carey et al., 2012; Cissner et al., 2013; Festinger et al., 2002; Jones, 2013; Marlowe et al., 2006, 2007; Mitchell et al., 2012; Rossman et al., 2011).⁴

Studies further reveal that consistent attendance by all team members at status hearings is associated with significantly better outcomes. A study of approximately seventy Drug Courts found that programs were 35% more cost-effective and 35% more effective at reducing crime when all team members—the judge, program coordinator, defense representative, prosecutor, probation officer, treatment representative, and law enforcement representative—attended status hearings regularly (Carey et al., 2012). When a treatment representative attended status hearings regularly, Drug Courts were nearly twice as effective at reducing crime and 80% more cost-effective, and when a representative from law enforcement attended hearings regularly, Drug Courts were over 80% more effective at reducing crime and 60% more cost-effective (Carey et al., 2008, 2012).

Although the judge typically controls most of the interactions during status hearings, observational studies reveal that other team members play an important role as well. Team members may report on participant progress, share their observations of participants, fill in missing information for the judge, offer praise and encouragement to participants, challenge inaccurate statements by participants, or make recommendations for suitable consequences to impose (Baker, 2013; Christie, 2014; Mackinem & Higgins, 2008; McPherson & Sauder, 2013; Portillo et al., 2013; Roper & Lessenger, 2007). Colloquially referred to as *courtroom as theater*, these interactions are often planned in advance during staffings to illustrate treatment-relevant concepts, prevent participants from fomenting disagreement among staff members, and demonstrate unity of purpose for the team as a whole (Satel, 1998; Tauber, 2011). In focus groups, participants rated interactions among staff during court sessions as informative and helpful to improving their performance (Goldkamp et al., 2002).

F. Team Training

Drug Courts represent a fundamentally new way of treating persons charged with drug-related offenses (Roper & Lessenger, 2007). Specialized knowledge and skills are required to implement these multifaceted programs effectively (Carey et al., 2012; Shaffer, 2010; Van Wormer, 2010). To be successful in their new roles, staff members require at least a journeyman's knowledge of best practices in a wide range of areas, including substance abuse and mental health treatment, complementary treatment and social services, behavior modification, community supervision, and drug and alcohol testing. Staff must also learn to perform their duties in a multidisciplinary environment, consistent with constitutional due process and the ethical mandates of their respective professions. These skills and knowledge sets are not taught in traditional law school, graduate school, or most continuing education programs (Berman & Feinblatt, 2005; Holland, 2010). Ongoing specialized training and supervision are needed for staff to achieve the goals of Drug Court and conduct themselves in an ethical, professional, and effective manner.

Preimplementation Trainings—In preimplementation trainings, staff meet for several days as a team to, among other things, develop a mission statement and goals and objectives for their program, learn from expert faculty about best practices in Drug Courts, and develop effective policies and procedures to govern their day-to-day operations (Hardin & Fox, 2011). A multisite study found that Drug Courts were nearly two and a half times more cost-effective and over 50% more effective at reducing recidivism when the teams participated in formal training prior to implementation (Carey et al., 2008, 2012). Drug Courts that

⁴ This finding assumes the Drug Court is serving the appropriate target population of high-risk and high-need participants [see Standard I, Target Population].

did not receive preimplementation training produced outcomes that were negligibly different from traditional criminal justice approaches (Carey et al., 2008).

Continuing Education Workshops—Continuing education workshops are commonly delivered as part of national, regional, or state Drug Court training conferences or in stand-alone seminars. These workshops provide experienced Drug Court professionals with up-to-date knowledge about new research findings on best practices in Drug Courts. Studies consistently find that annual attendance by staff at training workshops is associated with significantly better outcomes. A multisite study involving more than sixty Drug Courts found that annual attendance at training conferences was the greatest predictor of program effectiveness (Shaffer, 2006, 2010). Another large-scale study found that regular participation in continuing education workshops was the greatest predictor of a program’s adherence to the Drug Court model (Van Wormer, 2010). After taking continuing education into account, no other variable was independently or incrementally associated with adherence to the Drug Court model. This finding suggests that adherence to best practices may be mediated primarily through staff participation in continuing education workshops. The same study determined that regular attendance in continuing education workshops was also associated with better collaboration among Drug Court team members, increased job satisfaction by staff, greater perceived benefits of Drug Court, greater optimism about the effects of substance abuse treatment, and better perceived coordination between the criminal justice system and other social service and treatment systems (Van Wormer, 2010).

Tutorials for New Staff—Within five years, 30% to 60% of Drug Courts experience substantial turnover in key staff positions (Van Wormer, 2010). The highest turnover rates, commonly exceeding 50%, are among substance abuse and mental health treatment providers (Lutze & Van Wormer, 2007; McLellan et al., 2003; Taxman & Bouffard, 2003; Van Wormer, 2010). Evidence further reveals that staff turnover correlates significantly with downward drift in the quality of the services provided, meaning that services diverge increasingly from the Drug Court model as more staff positions turn over (Van Wormer, 2010).

Research has determined that Drug Courts are more effective when they provide introductory tutorials for new hires. A multisite study of approximately seventy Drug Courts found that programs were over 50% more effective at reducing recidivism when they routinely provided formal orientation training for new staff (Carey et al., 2012). Typically, the tutorials provide a “*Reader’s Digest*” orientation to the Ten Key Components of Drug Courts (NADCP, 1997) and a synopsis of best practices associated with each component. The tutorials are not intended to take the place of formal continuing education workshops, but serve rather as a stopgap measure to prevent acute disruption in services and degradation of outcomes. To maintain effective outcomes over time, recent hires should attend formal training workshops as soon as practicable after assuming their new positions. Given the powerful influence of staff training on Drug Court outcomes (Carey et al., 2012; Shaffer, 2006, 2010; Van Wormer, 2010), a firm commitment to ongoing professional education is key to maintaining the success and integrity of Drug Courts.

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IX. CENSUS AND CASELOADS

The Drug Court serves as many eligible individuals as practicable while maintaining continuous fidelity to best practice standards.

A. Drug Court Census

B. Supervision Caseloads

C. Clinician Caseloads

A. Drug Court Census

The Drug Court does not impose arbitrary restrictions on the number of participants it serves. The Drug Court census is predicated on local need, obtainable resources, and the program's ability to apply best practices. When the census reaches 125 active⁵ participants, program operations are monitored carefully to ensure they remain consistent with best practice standards. If evidence suggests some operations are drifting away from best practices, the team develops a remedial action plan and timetable to rectify the deficiencies and evaluates the success of the remedial actions.

B. Supervision Caseloads

Caseloads for probation officers or other professionals responsible for community supervision of participants must permit sufficient opportunities to monitor participant performance, apply effective behavioral consequences, and report pertinent compliance information during pre-court staff meetings and status hearings. When supervision caseloads exceed thirty active participants per supervision officer, program operations are monitored carefully to ensure supervision officers can evaluate participant performance accurately, share significant observations with team members, and complete other supervisory duties as assigned. Supervision caseloads do not exceed fifty active participants per supervision officer.

C. Clinician Caseloads

Caseloads for clinicians must permit sufficient opportunities to assess participant needs and deliver adequate and effective dosages of substance abuse treatment and indicated complementary services. Program operations are monitored carefully to ensure adequate services are delivered when caseloads exceed the following thresholds:

- 50 active participants for clinicians providing clinical case management⁶

⁵ Cases are considered to be active if participants are receiving treatment or supervision services from the Drug Court. Participants who have absconded from the program or are continuing on probation but no longer receiving Drug Court services are not considered active.

⁶ Clinical case management includes assessing participant needs, brokering referrals for indicated services, coordinating care between partner agencies, and reporting progress information to the Drug Court team (Braude, 2005; Monchick et al., 2006; Rodriguez, 2011). Clinical case managers may also represent treatment concerns during pre-court staff meetings and status

- 40 active participants for clinicians providing individual therapy or counseling
- 30 active participants for clinicians providing both clinical case management and individual therapy or counseling

COMMENTARY

A. Drug Court Census

Drug Courts serve fewer than 10% of adults in the criminal justice system in need of their services (Bhati et al., 2008; Huddleston & Marlowe, 2011). An important goal for the Drug Court field is to take Drug Courts to scale and serve every drug-addicted person in the criminal justice system who meets evidence-based eligibility criteria for the programs (Fox & Berman, 2002). Putting arbitrary restrictions on the size of the Drug Court census unnecessarily reduces the program's impact on public health and public safety.

Not all Drug Courts, however, may have adequate resources to increase capacity while maintaining fidelity to best practices. Surveys of judges and other criminal justice professionals consistently identify insufficient personnel and other resources as the principal barrier preventing Drug Courts from expanding to serve more people (Center for Court Innovation, n.d.; Farole, 2006, 2009; Farole et al., 2005; Huddleston & Marlowe, 2011). Resource limitations may put some Drug Courts in the challenging position of needing to choose between diluting their services to treat more people or turning away deserving individuals.

Evidence suggests expanding Drug Court capacity without sufficient resources can interfere with adherence to best practices. A multisite study of approximately seventy Drug Courts found a significant inverse correlation between the size of the Drug Court census and effects on criminal recidivism (Carey et al., 2008, 2012a). On average, programs evidenced a steep decline in effectiveness when the census exceeded approximately 125 participants. Drug Courts with fewer than 125 participants were over five times more effective at reducing recidivism than Drug Courts with more than 125 participants (Carey et al., 2012a).

Further analyses uncovered a likely explanation for this finding: Drug Courts with more than 125 participants were less likely to follow best practices than Drug Courts with fewer participants. Specifically, when the census exceeded 125 participants, the following was observed (Carey et al., 2012b):⁷

- Judges spent approximately half as much time interacting with participants in court.
- Team members were less likely to attend pre-court staff meetings.
- Treatment and law enforcement representatives were less likely to attend status hearings.
- Drug and alcohol testing occurred less frequently.
- Treatment agencies were less likely to communicate with the court about participant performance via email or other electronic means.
- Participants were treated by a large number of treatment agencies with divergent practices and expectations.
- Team members were less likely to receive training on Drug Court best practices.

hearings. Some court personnel or criminal justice professionals may be referred to as case managers or court case managers to be distinguished from clinical case managers. Court case managers may screen participants and refer them, when indicated, for more in-depth clinical assessments. These professionals do not provide clinical case management because they are not trained or qualified to administer clinical assessments, interpret assessment results, coordinate treatment delivery, or gauge treatment progress.

⁷ All comparisons statistically significant at $p < .05$.

These findings are merely correlations and do not prove that a large census produces poor outcomes. Most Drug Courts in the study were staffed by a single judge and a small team of roughly four to five other professionals overseeing a single court docket. Drug Courts can serve far more than 125 participants with effective results if the programs have sufficient personnel and resources to accommodate larger numbers of individuals. In fact, studies have reported positive outcomes for well-resourced Drug Courts serving more than 400 participants (Carey et al., 2012a; Cissner et al., 2013; Marlowe et al., 2008; Shaffer, 2010).

Nevertheless, the above results raise a red flag that as the census increases, Drug Courts may have greater difficulty delivering the quantity and quality of services required to achieve effective results. Therefore, when the Drug Court census reaches 125 active participants, this milestone should trigger a careful reexamination of the program's adherence to best practices. For example, staff should monitor Drug Court operations to ensure the judge is spending at least three minutes interacting with each participant in court [see Standard III, Roles and Responsibilities of the Judge], drug and alcohol testing is being performed randomly at least twice per week [see Standard VII, Drug and Alcohol Testing], team members are attending pre-court staff meetings and status hearings on a consistent basis [see Standard III and Standard VIII, Multidisciplinary Team], and team members are receiving up-to-date training on best practices [see Standards III and VIII]. If the results of this reexamination suggest some operations are drifting away from best practices, the team should develop a remedial action plan and timetable to rectify the deficiencies and evaluate the success of the remedial actions. For example, the Drug Court might need to hire additional staff to ensure it has manageable participant-to-staff caseloads, schedule status hearings on more days of the week, purchase more drug and alcohol tests, or schedule more continuing-education workshops for staff.

Studies have not determined whether censuses greater than 125 participants should trigger additional reexaminations of adherence to best practices. Until research addresses this question, at a minimum Drug Courts are advised to reexamine adherence to best practices when the census increases by successive increments of 125 participants.

B. Supervision Caseloads

In most Drug Courts, probation officers or pretrial services officers are responsible for supervising participants in the community; however, some Drug Courts may rely on law enforcement or specially trained court case managers to provide community supervision. Duties of the supervision officer may include performing drug and alcohol testing, conducting home and employment visits, enforcing curfews and geographic restrictions, and delivering cognitive-behavioral interventions designed to improve participants' problem-solving skills or alter dysfunctional criminal-thinking patterns (Harberts, 2011).

No study has examined the influence of supervision caseloads in Drug Courts. However, many studies have examined supervision caseloads in the context of adult probation. Early studies found that small probation caseloads were paradoxically associated with *increased* rates of technical violations and arrests for new offenses (Gendreau et al., 2000a; Petersilia, 1999; Turner et al., 1992). This counterintuitive finding was attributable to increased surveillance of the probationers coupled with a failure to apply evidence-based practices. Smaller caseloads led to greater detection of infractions, but most infractions received excessively punitive responses, such as probation revocations, rather than evidence-based treatment or gradually escalating incentives and sanctions (Andrews et al., 1990; Gendreau et al., 2000b; Hollin, 1999).

Recent studies have reported improved outcomes when reduced probation caseloads were combined with evidence-based cognitive-behavioral counseling, motivational interviewing, or gradually escalating incentives and sanctions (Jalbert & Rhodes, 2012; Jalbert et al., 2010, 2011; Paparozzi & Gendreau, 2005; Pearson & Harper, 1990; Worrall et al., 2004). Results of these newer studies confirm that detecting infractions alone is insufficient to improve outcomes. To achieve positive results, probation officers must respond to infractions and achievements by delivering effective behavioral contingencies (incentives and sanctions) and ensuring probationers receive effective and adequate evidence-based treatment and social services (Center for Effective Public Policy, 2014; Paparozzi & Hinzman, 2005; Skeem & Manchak, 2008).

Identifying optimal probation caseloads has been a challenging task. In 1990, the American Probation and Parole Association (APPA, 1991) issued caseload guidelines derived from expert consensus. The 1990

guidelines recommended caseloads of 30:1 for high-risk probationers who have a substantial likelihood of failing on probation or committing a new offense (Table 2). In 2006, the APPA guidelines were amended, in part, to add a new category for intensive supervised probation (ISP). ISP was designed for probationers who are both high risk and high need, meaning they pose a substantial risk of failing on probation and also have serious treatment or social-service needs (Petersilia, 1999). Because ISP and Drug Courts are both intended for high-risk and high-need individuals, recommendations for ISP may be particularly instructive for Drug Court best practices. Based on expert consensus, the 2006 APPA amendments recommended caseloads of 20:1 for high-risk and high-need probationers on ISP, and increased the recommended caseloads to 50:1 for moderate- and high-risk probationers who do not have serious treatment or social-service needs (Byrne, 2012; DeMichele, 2007).

TABLE 2	APPA* RECOMMENDED CASELOADS	
Probationer Risk and Need Level	1990 Guidelines	2006 Guidelines
ISP: [†] high risk and high need	NR [§]	20:1
High risk	30:1	50:1
Moderate risk	60:1	50:1
Low risk	120:1	200:1

*American Probation and Parole Association

Sources: APPA (1991); Byrne (2012); DeMichele (2007)

[†]Intensive supervised probation

[§]Not reported

Recent studies examined the effects of adhering to the 2006 APPA guidelines. A randomized experiment compared the services received and outcomes achieved when probation officers had reduced caseloads of approximately 50:1 for moderate and high-risk probationers as compared to typical probation caseloads of approximately 100:1 (Jalbert & Rhodes, 2012). Results confirmed that probationers on 50:1 caseloads received significantly more probation office sessions, field visits, employer contacts, telephone check-ins, and substance abuse and mental health treatment (Jalbert & Rhodes, 2012). As a consequence of receiving more services, they also had significantly better probation outcomes, including fewer positive drug tests and other technical violations (Jalbert & Rhodes, 2012). Probation officers with caseloads substantially above 50:1 had considerable difficulty accomplishing their core missions of monitoring probationers closely and reducing technical violations.

Another quasi-experimental study examined the effects of reducing caseloads from 50:1 to 30:1 for high-risk and high-need probationers on ISP (Jalbert et al., 2010). A 30:1 caseload is greater than the APPA recommended guideline of 20:1 for ISP, but is considerably smaller than typical probation caseloads of 100:1 (Bonta et al., 2008; Paparozzi & Hinzman, 2005) and recommended caseloads of 50:1 for most high-risk probationers (Byrne, 2012). Results confirmed that probationers on 30:1 caseloads had more frequent and longer contacts with their probation officers, and received more specialized services designed to reduce their risk to public safety, including behavior therapy, domestic-violence counseling, spousal-batterer interventions, and sex-offender treatment (Jalbert et al., 2010). Most striking, probationers on 30:1 caseloads had significantly lower recidivism rates lasting for at least two and a half years, including fewer new arrests for drug, property, and violent crimes (Jalbert et al., 2010).

Taken together, the weight of scientific evidence (Jalbert & Rhodes, 2012; Jalbert et al., 2011) and expert consensus (APPA, 1991; Byrne, 2012; DeMichele, 2007) suggests supervision officers are unlikely to manage high-risk cases effectively and reduce technical violations when their caseloads exceed 50:1. Supervision officers in Drug Courts are unlikely to accomplish their core functions of monitoring participants accurately, applying effective behavioral consequences, and sharing important compliance information with Drug Court team members if their caseloads exceed this critical threshold.

Research in ISP programs suggests long-term reductions in criminal recidivism are most likely to be achieved for high-risk and high-need participants when caseloads stay at or below 30:1 (Jalbert et al., 2010). Whether 30:1 caseloads are required similarly for Drug Courts is an open question. Drug Courts

include several components not encompassed by ISP, which may enhance the influence of supervision officers. For example, Drug Court participants are supervised and treated by a multidisciplinary team of professionals and attend status hearings in court on a frequent basis. Larger caseloads may be manageable for supervision officers in light of these additional service elements. Until research resolves the issue, Drug Courts are advised to monitor their operations carefully when caseloads for supervision officers exceed 30:1; caseloads should never exceed a 50:1 ratio. Assurance is needed that supervision officers can monitor participant performance effectively, contribute critical observations and information during pre-court staff meetings and status hearings, and complete other assigned duties such as performing drug and alcohol testing, conducting field visits, and delivering cognitive-behavioral criminal-thinking interventions.

Bear in mind these caseload guidelines assume the supervision officer is assigned principally to Drug Court and is not burdened substantially with other professional obligations. Smaller caseloads may be required if supervision officers are also managing caseloads outside of Drug Court or if they have supplementary administrative or managerial duties in addition to supervising Drug Court participants.

C. Clinician Caseloads

In Drug Courts, addiction counselors, social workers, psychologists, or clinical case managers are typically responsible for assessing participant needs, delivering or overseeing the delivery of treatment services, charting treatment progress, and reporting progress information to the Drug Court team (Lutze & Van Wormer, 2007; Shaffer, 2010; Van Wormer, 2010). Outcomes are significantly better in Drug Courts when participants meet individually with one of these clinicians on a weekly basis for at least the first phase of the program [see Standard V, Substance Abuse Treatment and Standard VI, Complementary Treatment and Social Services].

National studies of outpatient individual substance abuse treatment consistently find that the size of clinician caseloads is inversely correlated with patient outcomes and clinician job performance (Hser et al., 2001; McCaughrin & Price, 1992; Stewart et al., 2004; Vocisano et al., 2004; Woodward et al., 2006). As caseloads increase, patients receive fewer services, patients are more likely to abuse illicit substances, clinicians are more likely to behave punitively toward patients, and clinicians are more likely to report significant job burnout and dissatisfaction (King et al., 2004; Stewart et al., 2004). Comparable studies are lacking for residential substance abuse treatment and for group clinicians who deliver services to several participants simultaneously.

Determining appropriate caseloads for clinicians in Drug Courts depends largely on their role and the scope of their responsibilities:

- **Clinical Case Management Role**—Some clinicians in Drug Courts serve principally as clinical case managers, assessing participant needs, brokering referrals for services, and reporting progress information to the Drug Court team (Monchick et al., 2006). They may also represent treatment concerns during pre-court staff meetings and status hearings.
- **Treatment Provider Role**—Some clinicians serve principally as treatment providers, administering individual therapy or counseling and perhaps facilitating or cofacilitating group interventions (Cissner et al., 2013; Zweig et al., 2012). They may also provide or refer participants for indicated complementary services, such as mental health treatment or vocational counseling.
- **Combined Clinical Case Management and Treatment Provider Roles**—Some clinicians serve both clinical case management and treatment provider functions. In addition to providing individual therapy or counseling, they are responsible for assessing participant needs, referring participants for complementary services, coordinating care between multiple service providers, reporting progress to the Drug Court team, and representing treatment concerns during pre-court staff meetings and status hearings (Braude, 2005; Monchick et al., 2006).

National practitioner organizations have published broad caseload guidelines based in part on these professional roles and responsibilities (Case Management Society of America & National Association of Social Workers, 2008; North Carolina Administrative Office of the Courts, 2010; Rodriguez, 2011). These guidelines have not been validated empirically in terms of their effects on outcomes. Rather, they are

derived from expert consensus about heavy caseloads that are likely too large to deliver adequate services or that contribute to staff burnout and job dissatisfaction. The guidelines focus exclusively on individual counseling and clinical case management. Comparable guidelines for group counselors have not been published. Table 3 summarizes the consensus conclusions.

TABLE 3		CASELOAD GUIDELINES DERIVED FROM EXPERT CONSENSUS
Principal Role and Responsibilities	Caseload	Reference
Clinical case management	50:1 to 75:1	Rodriguez (2011)
Individual therapy or counseling	40:1 to 50:1	CMSA* & NASW† (2008) Hromco et al. (2003)
Combination of clinical case management and individual therapy or counseling	30:1	CMSA & NASW (2008) NCAOC§ (2010)

*Case Management Society of America

†National Association of Social Workers

§North Carolina Administrative Office of the Courts

To reiterate, these guidelines are derived from expert consensus and have not been validated against outcomes. Moreover, professional roles and responsibilities are rarely so clearly delineated in day-to-day Drug Court operations. Clinicians in Drug Courts may provide clinical case management for some participants and therapy or counseling for others, may have a mixture of individual and group treatment responsibilities, and may have other nonclinical duties, such as drug and alcohol testing, that reduce the time they have available for clinical assessment, treatment, or case management. Caseload expectations need to be adjusted in light of actual job responsibilities.

Nevertheless, these guidelines should serve as broad milestones to alert Drug Courts to the possibility of clinician overload and the need to audit their operations to ensure adequate services are being delivered. Because Drug Courts serve high-risk and high-need individuals, programs are advised to reexamine adherence to best practices when clinician caseloads reach the lowest ratios reported in Table 3. For example, when clinical case management caseloads exceed 50:1, individual counseling caseloads exceed 40:1, or combined caseloads exceed 30:1, staff should monitor Drug Court operations to ensure participants are being assessed appropriately for risk and need [see Standard I, Target Population], participants are meeting individually with a clinician on a weekly basis for at least the first phase of treatment [see Standard V, Substance Abuse Treatment and Standard VI, Complementary Treatment and Social Services], participants are receiving at least 200 hours of cognitive-behavioral treatment [see Standard V], and clinicians are providing reliable and timely progress information to the Drug Court team [see Standard VIII, Multidisciplinary Team]. Drug Courts are unlikely to achieve the goals of rehabilitating participants and reducing crime if clinicians are spread too thin to assess and meet participants' service needs.

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X. MONITORING AND EVALUATION

The Drug Court routinely monitors its adherence to best practice standards and employs scientifically valid and reliable procedures to evaluate its effectiveness.⁸

- A. Adherence to Best Practices**
- B. In-Program Outcomes**
- C. Criminal Recidivism**
- D. Independent Evaluations**
- E. Historically Disadvantaged Groups**
- F. Electronic Database**
- G. Timely and Reliable Data Entry**
- H. Intent-to-Treat Analyses**
- I. Comparison Groups**
- J. Time at Risk**

A. Adherence to Best Practices

The Drug Court monitors its adherence to best practice standards on at least an annual basis, develops a remedial action plan and timetable to rectify deficiencies, and examines the success of the remedial actions. Outcome evaluations describe the effectiveness of the Drug Court in the context of its adherence to best practices.

B. In-Program Outcomes

The Drug Court continually monitors participant outcomes during enrollment in the program, including attendance at scheduled appointments, drug and alcohol test results, graduation rates, lengths of stay, and in-program technical violations⁹ and new arrests.

C. Criminal Recidivism

Where such information is available, new arrests, new convictions, and new incarcerations are monitored for at least three years following each participant's entry into the Drug Court. Offenses are categorized according to the level (felony, misdemeanor, or summary offense) and nature (e.g., person, property, drug, or traffic offense) of the crime involved.

⁸ Herein, monitoring refers to periodic descriptions of the services delivered and outcomes achieved in a Drug Court without inferring a causal relationship between the services and outcomes. An evaluation includes a comparison condition and other scientific procedures designed to attribute outcomes to the effects of the Drug Court. Most Drug Courts are capable of monitoring their services and outcomes but may require expert consultation to evaluate the causal effects of their program.

⁹ A *technical violation* refers to a violation of a court order that does not constitute a crime per se. For example, drinking alcohol is legal for most adults but is usually a technical violation in a Drug Court.

D. Independent Evaluations

A skilled and independent evaluator examines the Drug Court's adherence to best practices and participant outcomes no less frequently than every five years. The Drug Court develops a remedial action plan and timetable to implement recommendations from the evaluator to improve the program's adherence to best practices.

E. Historically Disadvantaged Groups

The Drug Court continually monitors admission rates, services delivered, and outcomes achieved for members of historically disadvantaged groups who are represented in the Drug Court population. The Drug Court develops a remedial action plan and timetable to correct disparities and examines the success of the remedial actions [see also Standard II, Historically Disadvantaged Groups].

F. Electronic Database

Information relating to the services provided and participants' in-program performance is entered into an electronic database. Statistical summaries from the database provide staff with real-time information concerning the Drug Court's adherence to best practices and in-program outcomes.

G. Timely and Reliable Data Entry

Staff members are required to record information concerning the provision of services and in-program outcomes within forty-eight hours of the respective events. Timely and reliable data entry is required of each staff member and is a basis for evaluating staff job performance.

H. Intent-to-Treat Analyses

Outcomes are examined for all eligible participants who entered the Drug Court regardless of whether they graduated, withdrew, or were terminated from the program.

I. Comparison Groups

participants in the other group, the length of time participants were detained or incarcerated is accounted for statistically in outcome comparisons.

COMMENTARY

A. Adherence to Best Practices

Adherence to best practices is generally poor in most sectors of the criminal justice and substance abuse treatment systems (Friedmann et al., 2007; Henderson et al., 2007; McLellan et al., 2003; Taxman et al., 2007). Programs infrequently deliver services that are proven to be effective and commonly deliver services which have not been subjected to careful scientific scrutiny. Over time, the quality and quantity of the services provided may decline precipitously (Etheridge et al., 1995; Van Wormer, 2010). The best way for a Drug Court to guard against these prevailing destructive pressures is to monitor its operations routinely, compare its performance to established benchmarks, and seek to align itself continually with best practices. Not knowing whether one's Drug Court is in compliance with best practices makes it highly unlikely that needed improvements will be recognized and implemented; therefore, evaluating a Drug Court's adherence to best practice standards is, itself, a best practice.

Studies reveal that Drug Courts are significantly more likely to deliver effective services and produce positive outcomes when they hold themselves accountable for meeting empirically validated benchmarks for success. A multisite study involving approximately seventy Drug Courts found that programs had more than twice the impact on crime and were more than twice as cost-effective when they monitored their operations on a consistent basis, reviewed the findings as a team, and modified their policies and procedures accordingly (Carey et al., 2008, 2012).

Like many complex service organizations, Drug Courts are highly susceptible to *drift*, in which the quality of their services may decline appreciably over time (Van Wormer, 2010). Management strategies such as continuous performance improvement (CPI), continuous quality improvement (CQI), and managing for results (MFR) are designed to avoid drift and enhance a program's adoption of best practices. Each of these management strategies emphasizes continual self-monitoring and rapid-cycle testing. This process involves collecting real-time information about a program's operations and outcomes, feeding that information back to key staff members and decision makers on a routine basis, and implementing and evaluating remedial action plans where indicated. Research consistently shows that continual self-monitoring and rapid-cycle testing are critical elements for improving outcomes and increasing adoption of best practices in the health care and criminal justice systems (Damschroder et al., 2009; Rudes et al., 2013; Taxman & Belenko, 2013). These strategies are essential for programs that require cross collaboration and interdisciplinary communication among multiple service agencies, including Drug Courts (Bryson et al., 2006; Wexler et al., 2012).

Studies have not determined how frequently programs should review performance information and implement and evaluate self-corrective measures. Common practice among successful organizations is to collect performance data continually and meet at least annually as a team to review the information and take self-corrective measures (Carey et al., 2012; Rudes et al., 2013; Taxman & Belenko, 2013).

Reporting outcomes from Drug Courts without placing those findings into context by describing the quality of the programs is no longer enough. Meta-analyses (Aos et al., 2006; Latimer et al., 2006; Lowenkamp et al., 2005; Mitchell et al., 2012; Shaffer, 2010; Wilson et al., 2006) and large-scale multisite studies (Rossman et al., 2011) have already clearly established that Drug Courts reduce crime by approximately 8% to 14% on average. These averages, derived from evaluations of more than 100 Drug Courts, mask a great deal of variability between programs. Some Drug Courts reduce crime by more than 50%, others have no impact on crime, and still others increase crime rates in their communities (Carey et al., 2012; Carey & Waller, 2011; Cissner et al., 2013; Downey & Roman, 2010; Government Accountability Office, 2011; Mitchell et al., 2012; Shaffer, 2010). The important question is no longer whether Drug Courts can work,

but rather how they work and what services contribute to better outcomes (Marlowe et al., 2006). Understanding what distinguishes effective Drug Courts from ineffective and harmful Drug Courts is now an essential goal for the field. Unless evaluators describe each Drug Court's adherence to best practices, there is no way to place that program's outcomes in context or interpret the significance of the findings.

B. In-Program Outcomes

One of the primary aims of a Drug Court is to rehabilitate seriously addicted individuals, which means that retaining participants in treatment, reducing drug and alcohol use, and helping participants to complete treatment successfully are important indicators of short-term progress. However, policymakers, the public, and other stakeholders are likely to judge the merits of a Drug Court by how well it reduces crime, incarceration rates, and taxpayer expenditures. Therefore, Drug Courts need to measure in-program outcomes that not only reflect clinical progress, but are also significant predictors of postprogram criminal recidivism and other long-term outcomes.

At minimal cost and effort, Drug Courts can evaluate short-term outcomes while participants are enrolled in the program. These short-term outcomes provide significant information about participants' clinical progress and the likely long-term impacts of the Drug Court on public health and public safety. Studies have consistently determined that postprogram recidivism is reduced significantly when participants attend more frequent treatment and probation sessions, provide fewer drug-positive urine tests, remain in the program for longer periods of time, have fewer in-program technical violations and arrests for new crimes, and satisfy other conditions for graduation (Gifford et al., 2014; Gottfredson et al., 2007, 2008; Huebner & Cobbina, 2007; Jones & Kemp, 2011; Peters et al., 2002). Drug Courts should, therefore, monitor and report on these in-program outcomes routinely during the course of their operations.

Several resources are available to help Drug Courts define and calculate performance measures of in-program outcomes (Berman et al., 2007; Heck, 2006; Marlowe, in press; Peters, 1996; Rubio et al., 2008a). In 2006, NADCP convened leading Drug Court researchers and evaluators to form the National Research Advisory Committee (NRAC). One goal of this committee was to define a core data set of in-program performance measures for adult Drug Courts (Heck, 2006). NRAC selected measures that are simple and inexpensive to track and evaluate and proven to predict long-term outcomes. These performance measures include the following:

- *Retention*—the number of participants who completed the Drug Court divided by the number who entered the program
- *Sobriety*—the number of negative drug and alcohol tests divided by the total number of tests performed
- *Recidivism*—the number of participants arrested for a new crime divided by the number who entered the program, and the number of participants adjudicated officially for a technical violation divided by the number who entered the program
- *Units of Service*—the numbers of treatment sessions, probation sessions, and court hearings attended
- *Length of Stay*—the number of days from entry to discharge or the participant's last in-person contact with staff

Longer lists of performance measures addressing a wide range of outcomes in Drug Courts and other problem-solving courts have been published by expert organizations including the National Center for State Courts (Rubio et al., 2008a; Waters et al., 2010), the Center for Court Innovation (Rempel, 2006, 2007), American University (Peters, 1996), the Organization of American States (Marlowe, in press), the National Center for DWI Courts (Marlowe, 2010), and the National Institute of Justice (NIJ, 2010). Drug Courts are advised to consult these and other resources for further information on how to calculate and interpret additional performance measures for their evaluations.

C. Criminal Recidivism

For many policymakers and members of the public, reducing criminal recidivism is one of the primary aims of a Drug Court. Recidivism is defined as any return to criminal activity after the participant entered the Drug Court. Recidivism does not include crimes that occurred before the participant entered Drug Court even if those crimes are charged or prosecuted after entry.

Recidivism is measured most commonly by new arrests, new convictions, or new incarcerations occurring over a two- or three-year period (Carey et al., 2012; King & Elderbroom, 2014; Rempel, 2006). For example, the Bureau of Justice Statistics (BJS) tracks new arrests, convictions, and incarcerations occurring within three years of the date that state and federal inmates are released from jail or prison (Durose et al., 2014).

Based on scientific considerations, evaluators should follow participants for at least three years, and ideally up to five years, from the date of entry into the Drug Court or from the date of the arrest or technical violation that made the individual eligible for Drug Court. The date of entry should be the *latest* start date for the evaluation because that is when the Drug Court becomes capable of influencing participant behavior directly.

Starting from the date of arrest or technical violation takes into account the potential impact of delays in admitting participants to Drug Court. The sooner participants enter Drug Court after an arrest or probation violation, the better the results (Carey et al., 2008, 2012); therefore, evaluators may wish to examine how delayed entry affects outcomes. However, because Drug Courts cannot always control what transpires before participants enter the Drug Court program, attributing to the Drug Court any recidivism occurring before entry may not fairly represent the Drug Courts' effects on recidivism. Starting from the date of entry ensures recidivism may be attributed fairly to the effects of the Drug Court. No one answer fully addresses the issues surrounding selection of a start date for evaluation; therefore, evaluators should state clearly what start date was selected and the rationale for choosing that start date.

Rates of criminal recidivism among drug-involved offenders become relatively stable after approximately three to five years (King & Elderbroom, 2014). After three years, statistically significant between-group differences in recidivism are likely to remain significant going forward (e.g., Knight et al., 1999; Martin et al., 1999; Wexler et al., 1999). For example, if Drug Court participants have significantly lower rearrest rates than comparison group subjects after three years, this difference is likely (although not guaranteed) to remain significant after an additional two years (DeVall et al., 2015). After five years, recidivism rates tend to reach a plateau, meaning that most (but not all) participants who will recidivate have likely done so by then (e.g., Gossop et al., 2005; Inciardi et al., 2004; Olson & Lurigio, 2014).

Importantly, these findings do not suggest Drug Courts must wait three to five years before reporting recidivism outcomes. Recidivism occurring during enrollment and shortly after discharge from Drug Court may be of considerable interest to practitioners, policymakers, and other stakeholders. However, implying that recidivism rates occurring within the first two years are likely to reflect the long-term effects of a Drug Court is inappropriate. Evaluators should state clearly that such recidivism rates are preliminary and likely to increase over time.

No one basis exists for deciding whether new arrests, new convictions, or new incarcerations are likely to be the most valid or informative indicator of recidivism. As discussed below, each measure has advantages and disadvantages that the evaluator must take into account. Because no one measure is clearly superior to another, whenever possible evaluators are advised to report all three measures of recidivism, discuss the implications and limitations of each, or indicate why a particular measure is not being reported.

Analyzing new arrests as a measure of criminal recidivism provides at least two advantages. First, arrests are often substantially closer in time to the alleged offense than convictions. Resolving a criminal case and determining guilt or innocence may take months or years. Evaluators can usually report arrest outcomes in much less time than waiting for lengthy legal proceedings to resolve. Second, criminal cases are often dismissed or pled down to a lesser charge for reasons having little to do with factual guilt, such as

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insufficient evidence or plea bargains. As a result, the absence of a conviction or conviction on a lesser charge may not reflect the offense that occurred.

However, some individuals are arrested for crimes they did not commit. This fact may lead to an overestimation of the true level of criminal recidivism. Relying on conviction data rather than arrest data may provide greater assurances that the crimes did, in fact, occur.

Incarceration has substantial cost impacts that may far exceed those of arrests and convictions. A day in jail or prison can cost between five and twenty times more than a day on probation or in community-based treatment (Belenko et al., 2005; Zarkin et al., 2012). Evaluators typically distinguish between incarceration that occurred while participants were enrolled in the Drug Court and incarceration that occurred after discharge. In-program incarceration often reflects brief jail sanctions that may be imposed for misconduct in the program, whereas postprogram incarceration typically reflects pretrial detention for new charges, sentences for new charges, or (for terminated participants) sentencing on the original charge that led to participation in Drug Court. In cost evaluations, in-program jail sanctions are typically counted as an investment cost for the Drug Court whereas postprogram detention is typically counted as an outcome cost (Carey et al., 2012).

Evaluators must also consider the timeliness and accuracy of information contained in criminal justice databases. In some jurisdictions, arrest data may be recorded in a more timely and faithful manner than conviction or incarceration data. Evaluators must familiarize themselves with how and when information is entered into national, state, and local criminal justice records and should describe clearly in their evaluation reports any limitations that may relate to the accuracy or timeliness of the data.

Self-report information could potentially provide the most accurate assessment of criminal recidivism because it does not require detection or prosecution by law enforcement. Because many crimes are unreported by victims and undetected by the authorities (Truman & Langton, 2014), arrest and conviction data may underestimate true levels of criminal activity. For obvious reasons, however, individuals cannot be relied upon to acknowledge their crimes unless they receive strict assurances that the information will be kept confidential and will not be used against them in a criminal proceeding. Drug Courts will typically be required to hire an independent evaluator who has no connection to the court or criminal justice system to confidentially survey participants. This method is likely to be prohibitively costly for many Drug Courts, which explains why it has rarely been employed with the notable exception of one highly funded national study (Rossman et al., 2011).

Whether measured by arrests, convictions, or incarcerations, categorizing recidivism according to the level (i.e., felony, misdemeanor, or summary offense) and nature (e.g., drug offenses, property and theft offenses, violent offenses, technical violations, prostitution, and traffic offenses) of the crimes involved is highly informative and necessary. Different categories of crime can have very different implications for public safety and cost. For example, violent offenses may have serious victimization costs and may result in substantial jail or prison sentences, whereas drug possession may not involve an identifiable victim and is more likely to receive a less costly probation sentence (Zarkin et al., 2012).

As a final note, not all Drug Courts have reasonable access to data on new arrests, convictions, or incarcerations occurring after participants have been discharged from the program. In some jurisdictions, these records may be in the possession of other executive agencies, such as the police department or department of corrections, and the Drug Court may not be entitled to the information. Under such circumstances, Drug Courts should make every effort to negotiate access to the data, but of course, Drug Courts cannot be held accountable for reporting information beyond their reach.

D. Independent Evaluations

In addition to monitoring their own performance, Drug Courts benefit greatly from having an independent evaluator examine their program and issue recommendations to improve their adherence to best practices. Drug Courts that engaged an independent evaluator and implemented at least some of the evaluator's recommendations were determined in one multisite study to be twice as cost-effective and nearly twice as

effective at reducing crime as Drug Courts that did not engage an independent evaluator (Carey et al., 2008, 2012).

Drug Courts benefit from an independent evaluation for several reasons. Every program has blind spots that prevent staff from recognizing their own shortcomings. Some team members, such as the judge, may have more social influence or power than others, making it difficult for some team members to call attention to problems in court or during team meetings. Drug Courts also operate in a political environment and staff may be hesitant to criticize local practices for fear of reprisal. An independent evaluator from another jurisdiction can usually offer frank criticisms of current practices with less fear of repercussions (Heck & Thanner, 2006).

Although most Drug Courts are capable of keeping descriptive statistics about their program, considerably more expertise is required to perform *inferential analyses*, which compare Drug Court outcomes to those of a comparison group. Controlling statistically for preexisting group differences that could bias one's results is often necessary. For example, if Drug Court participants had fewer previous convictions than comparison subjects before entering the study, better outcomes for the Drug Court might simply reflect the fact that it treated a less severe population. Evaluators must take numerous scientific matters into consideration and may need to apply several levels of statistical corrections to produce valid and reliable results.

Studies also reveal that participant perceptions are often highly predictive of outcomes in Drug Courts. For example, perceptions concerning the procedural fairness of the program (Burke, 2010; McIvor, 2009), the manner in which incentives and sanctions are delivered (Goldkamp et al., 2002; Harrell & Roman, 2001; Marlowe et al., 2005), and the quality of the treatment services provided (Turner et al., 1999) are often predictive of recidivism and correlate significantly with adherence to best practices. Needless to say, participants are more likely to be forthright with an independent evaluator about their perceptions of the Drug Court than with staff members who control their fate in the criminal justice system.

Studies have not determined how frequently Drug Courts should be evaluated by an independent investigator. Generally speaking, a new evaluation should be performed whenever a program or the environment within which it operates changes substantially. Staff turnover and evidence of drift from the intended model are critical events that call for a new evaluation (Yeaton & Camberg, 1997). Evidence suggests that staff turnover and model drift occur within five-year intervals in Drug Courts. Within five years, between roughly 30% and 60% of Drug Courts experience substantial turnover in key staff positions (Van Wormer, 2010). The highest turnover rates, commonly exceeding 50%, are among substance abuse and mental health treatment providers (Lutze & Van Wormer, 2007; McLellan et al., 2003; Taxman & Bouffard, 2003; Van Wormer, 2010). Evidence further reveals that staff turnover correlates significantly with drift in the quality of the services provided (Van Wormer, 2010). Therefore, five years is a reasonable outside estimate of how frequently Drug Courts should be evaluated independently. If resources allow, Drug Courts should engage independent evaluators at more frequent intervals to detect drift readily and prevent services from worsening with time.

Drug Courts need to select competent evaluators. The first step in selecting a competent evaluator is to request recommendations from other Drug Courts and national organizations that are familiar with Drug Court operations and research. Senior staff at NADCP and NDCI are familiar with the evaluation literature and the skill sets of dozens of evaluators nationally. When selecting an evaluator, review prior evaluation reports, especially those involving Drug Courts or other problem-solving courts. If prior evaluations failed to follow the practices described herein, consider selecting another evaluator who has demonstrated expertise in applying best practices related to Drug Court program evaluations. One of the most important questions to consider when reviewing prior evaluations is whether the report recommended concrete actions the Drug Court could take to enhance its adherence to best practices and improve its outcomes. The most effective evaluators are aware of the literature on best practices, measure Drug Court practices against established performance benchmarks, and promote useful strategies to improve each program's operations and results.

Many Drug Courts do not have sufficient resources to hire independent evaluators. One way to address this problem is to contact local colleges or universities to determine whether graduate or undergraduate students may be interested in evaluating the Drug Court as part of a thesis, dissertation, or capstone project. Because

such projects require close supervision from senior academic faculty, the Drug Court can receive high-level research expertise at minimal or no cost. Moreover, students are likely to be highly motivated to complete the evaluation successfully because their academic degree and standing depends on it.

E. Historically Disadvantaged Groups

The term *historically disadvantaged groups* refers to socio-demographic groups that have historically experienced sustained discrimination or reduced social opportunities due to their race, ethnicity, gender, sexual orientation, sexual identity, physical or mental disability, religion, or socioeconomic status. Best practices for ensuring equivalent treatment of historically disadvantaged groups in Drug Courts are described in Standard II, Historically Disadvantaged Groups.

Evidence suggests racial and ethnic minority individuals are underrepresented in some Drug Courts and may have lower graduation rates than other participants [see Commentary in Standard II, Historically Disadvantaged Groups]. Drug Courts have an affirmative obligation to determine whether racial and ethnic minority individuals and members of other historically disadvantaged groups are being disproportionately burdened or excluded from their programs; and if so, to take reasonable corrective measures to rectify the problem and evaluate the success of the corrective actions [see Standard II]. Not knowing whether one's Drug Court is disproportionately burdening disadvantaged groups is itself a violation of best practice standards (Marlowe, 2013).

Studies have not determined how frequently Drug Courts should review performance information for members of historically disadvantaged groups. Consistent with the general literature on CPI, CQI and MFR, the Drug Court team should review performance information at least annually and implement and evaluate self-corrective measures on a rapid-cycle basis (Rudes et al., 2013; Wexler et al., 2012).

A number of resources are available to help Drug Courts identify and rectify disparate impacts for historically disadvantaged groups (e.g., Casey et al., 2012; Rubio et al., 2008b; Yu et al., 2009). Seasoned evaluators and university faculty are likely to be familiar with this literature and to know how to perform these types of analyses. Many analyses, such as comparing graduation rates between different racial groups, are relatively simple and straightforward to perform. Other analyses, such as determining whether disadvantaged groups have equivalent access to Drug Court, are considerably more difficult. Many Drug Courts may not have adequate information about the relevant arrestee population to determine whether disadvantaged groups are gaining access to the Drug Court at equivalent rates. For example, information might not be available to determine what proportion of racial-minority arrestees have serious drug problems and are therefore eligible for participation in Drug Court. The primary challenge for such Drug Courts may be to gain better access to a wider range of information on the arrestee population, and as a practical matter, such analyses may be beyond the ability and expertise of some programs to accomplish.

F. Electronic Database

Paper files have minimal value for conducting program evaluations. Evaluators are typically required to extract information from handwritten notes and progress reports that are difficult to read, contain contradictory information, and have numerous missing entries. As a consequence, many evaluations are completed months or years after the fact when the results may no longer reflect what is occurring in the program. Such evaluations often contain so many gaps or caveats in the data that the conclusions which may be drawn are tentative at best.

Drug Courts are approximately 65% more cost-effective when they enter standardized information concerning their services and outcomes into an electronic management information system (MIS), which is capable of generating automated summary reports (Carey et al., 2008, 2012). The cost of purchasing an MIS is offset many times over by providing greater efficiencies in operations and yielding the type of performance feedback that is necessary to continually improve and fine-tune one's Drug Court program.

Appendix E provides examples of MISs that have been developed for use in Drug Court evaluations. Some of the older and less sophisticated systems can be obtained free of charge. For example, the Buffalo System (so named because it was developed in a Drug Court in Buffalo, New York) is a Microsoft Access database

that can be obtained at no cost by contacting NADCP. Newer systems must be purchased or licensed, but are more likely to be web-based and can be accessed simultaneously by multiple users and agencies. Allowing multiple agencies to use the same MIS, each with its own secured and encrypted access, can spread the cost of the system across several budgets. Newer systems are also more likely to have preprogrammed analytic reports that provide important summary information for staff at the push of a button. Finally, newer systems are more likely to include a data-extraction tool. A data-extraction tool allows information to be imported readily into a statistical program, such as SAS or SPSS, which skilled evaluators then can use to conduct sophisticated statistical analyses.

G. Timely and Reliable Data Entry

The biggest threat to a valid program evaluation is poor data entry by staff. The adage “garbage in/garbage out” is particularly apt in this regard. If staff members do not accurately record what occurred, no amount of scientific expertise or sophisticated statistical adjustments can produce valid findings.

The best time to record information about services and events is when they occur. For example, staff members should enter attendance information into an MIS or written log during court hearings and treatment sessions. This is referred to as *real-time recording*. The typical staff person in a Drug Court is responsible for dozens of participants and each participant has multiple obligations in the program, such as appearing at court hearings, attending treatment sessions, and delivering urine specimens. Only the rare staff person can recall accurately what events transpired or should have transpired days or weeks in the past. Attempting to reconstruct events from memory is likely to introduce unacceptable error into a program evaluation.

Data should ordinarily be recorded within no more than forty-eight hours of the respective events. Medicare, for instance, requires physicians to document services within a “reasonable time frame,” defined as twenty-four to forty-eight hours (Pelaia, n.d.). After forty-eight hours, errors in data entry have been shown to increase significantly. After one week, information is so likely to be inaccurate that it may be better to leave the data as missing than attempt to fill in gaps from faulty memory (Marlowe, 2010).

Staff members who are persistently tardy when entering data pose a serious threat to the integrity of a Drug Court. Not only are evaluation results unlikely to be accurate, but those same staff persons are unlikely to be delivering appropriate services. Good-quality treatment and supervision require staff to monitor participant behavior vigilantly, record performance information in a timely and actionable fashion, and adjust services and consequences accordingly. Failing to record performance information in a timely and reliable manner undermines the quality and effectiveness of a Drug Court and seriously jeopardizes participant care.

H. Intent-to-Treat Analyses

A serious error in some Drug Court evaluations is to examine outcomes only for participants who graduated successfully from the program. The logic for performing such an analysis is understandable. Evaluators are often interested in learning what happens to individuals who received all of the services the program has to offer. If individuals who dropped out or were terminated prematurely from the Drug Court are included in the analyses, the results will be influenced by persons who did not receive all of the intended services.

Although this reasoning might seem logical, it is scientifically flawed (Heck, 2006; Heck & Roussell, 2007; Marlowe, 2010, in press; Peters, 1996; Rempel, 2006, 2007). Outcomes must be examined for all eligible individuals who participated in the Drug Court regardless of whether they graduated, were terminated, or withdrew from the program. This is referred to as an *intent-to-treat analysis* because it examines outcomes for all individuals whom the program initially set out to treat. Reporting outcomes for graduates alone is not appropriate because such an analysis unfairly and falsely inflates the apparent success of the program. For example, individuals who graduated from the Drug Court are more likely than terminated participants to have entered the program with less severe drug or alcohol problems, less severe criminal propensities, higher motivation for change, or better social supports. As a result, they might have been less likely to commit future offenses or relapse to substance abuse regardless of the services they received in Drug Court.

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This issue is particularly important when outcomes are contrasted against those of a comparison sample, such as probationers. Selecting the most successful Drug Court cases and comparing their outcomes to all of the probationers unfairly skews the results in favor of the Drug Court. It is akin to selecting the A+ students from one classroom, comparing their scores on a test to those of all of the students in a second classroom, and then concluding the first class had a better teacher. Such a comparison would clearly be slanted unfairly in favor of the first teacher.

This is not to suggest that outcomes for graduates are of no interest. Drug Courts may, indeed, want to know what happens to individuals who receive all of the services in the program. This, however, should be a *secondary analysis* that is performed after the intent-to-treat analysis has shown positive results. If it is first determined that the Drug Court achieved significantly improved outcomes on an intent-to-treat basis, it may then be appropriate to proceed further and determine whether outcomes were even better for the graduates. If the intent-to-treat analysis is not significant, then it is not acceptable to move on to evaluate outcomes for graduates alone.

Importantly, if secondary analyses are performed on Drug Court graduates, then the comparison sample should also comprise successful completers. For example, outcomes for Drug Court graduates should be compared to those of probationers who satisfied the conditions of probation. Comparing outcomes for Drug Court graduates to all probationers, including probation failures, would unfairly favor the Drug Court.

The only exception to an intent-to-treat analysis is for what are sometimes referred to as *neutral discharges*. Some Drug Courts assign a neutral discharge to participants who are withdrawn from the program for reasons beyond the control of the participant and the program. A neutral discharge is assigned most commonly when the Drug Court discovers a participant was admitted to the program erroneously. For example, a participant might need to be withdrawn from Drug Court if he or she had a prior conviction that precluded eligibility for the Drug Court or resided in a judicial district that was not within the jurisdictional boundaries of the Drug Court. A neutral discharge may also be assigned to participants who are withdrawn from the program because they enlisted in the military or moved out of the jurisdiction with the court's permission. A neutral discharge should never be assigned to cases in which termination was related to a participant's performance in Drug Court.

I. Comparison Groups

The mere fact that individuals perform well after participating in Drug Court does not prove the Drug Court was responsible for their favorable outcomes. Those same individuals might have functioned just as well if they had never entered Drug Court. To examine the important question of causality, the performance of Drug Court participants must be compared against that of an equivalent and unbiased comparison group. Comparing what happened in the Drug Court to what would most likely have happened if the Drug Court did not exist is referred to as testing the *counterfactual hypothesis*, or the possibility that the Drug Court was ineffective (Popper, 1959).

Some comparison groups are reasonably unbiased and can yield a fair and accurate assessment of what would most likely have occurred without the Drug Court. Others, however, may be systematically biased in such a manner as to make the Drug Court look better or worse than it deserves. This may lead to the unwarranted conclusion that the Drug Court was effective or ineffective when, in fact, the reverse could be true.

Random Assignment—The strongest inference of causality may be reached when eligible individuals are randomly assigned either to the Drug Court or to a comparison group. Random assignment provides the greatest assurance that the groups started out with an equal chance of success; therefore, better outcomes for one group can be confidently attributed to the effects of the program (Campbell & Stanley, 1963; Farrington, 2003; Farrington & Welsh, 2005; National Research Council, 2001; Telep et al., 2015). Even when an evaluator employs random assignment, there is still the possibility (albeit a greatly diminished possibility) that the groups differed on important dimensions from the outset. This possibility requires the evaluator to perform a confirmation of the randomization procedure. The evaluator will need to check for preexisting differences between the groups that could have affected the results. If the groups differed significantly on variables that are correlated with outcomes (such as the severity of participants' criminal

histories or drug problems), the evaluator might employ statistical procedures to adjust for those differences and obtain defensible results.

As a practical matter, conducting random assignment is often very difficult in Drug Courts. Some staff members may have ethical objections against denying potentially effective services to eligible individuals. Moreover, some Drug Courts may have difficulty filling their slots and may not wish to turn away eligible individuals. The evaluator will also need to gain approval and buy-in for random assignment from numerous professionals and agencies, including the court, prosecution, and defense counsel. Finally, random assignment usually requires implementation of ethical safeguards (National Research Council, 2001). For example, participants may need to provide informed consent to random assignment, and an independent ethics review board may need to oversee the safety and fairness of the study. Local colleges and universities often have institutional review boards (IRBs) or data and safety monitoring boards (DSMBs) which have the authority and expertise to provide ethical oversight for randomized studies.

Random assignment poses far fewer challenges if a Drug Court has insufficient capacity to treat many individuals who would otherwise be eligible for its services. If many eligible people must be turned away, then it would arguably be fairest to select participants randomly rather than allow staff members to pick and choose who gets into the program. Under such circumstances, random assignment may provide the best protection against unfair discrimination and unconscious bias (National Research Council, 2001). In fact, a number of Drug Court studies have used random assignment successfully in light of insufficient program capacity (e.g., Gottfredson et al., 2003; Jones, 2011; Turner et al., 1999).

Quasi-Experimental Comparison Group—In many Drug Courts, engaging in random assignment is simply impractical. The next best approach is to use a quasi-experimental comparison group (Campbell & Stanley, 1963). This refers to individuals who were eligible for the Drug Court but did not enter for reasons that are unlikely to have influenced their outcomes. Perhaps the best example is individuals who were eligible for and willing to enter the Drug Court, but were denied access because there were no empty slots available. This is referred to as a *wait-list comparison group*. The mere happenstance that the Drug Court was full is unlikely to have led to the systematic exclusion of individuals who had more severe problems or poorer prognoses to begin with, and therefore is unlikely to bias the results.

Less optimal, but still potentially acceptable, quasi-experimental comparison groups include individuals who would have been eligible for the Drug Court but were arrested in the year or so before the Drug Court was established, or were arrested in an immediately adjacent county that does not have a Drug Court (Heck, 2006; Heck & Roussell, 2007; Marlowe, 2010, in press; Peters, 1996). Because these individuals were arrested at an earlier point in time or in a different geographic region than the Drug Court participants, such comparison groups might still be different enough from the Drug Court group to bias the results. For example, socioeconomic conditions might differ significantly between neighboring communities, or law enforcement practices might change from year to year. The likelihood of this occurring, however, is usually not substantial and these may be the only practical comparison conditions that can be used for many Drug Court evaluations.

When using a quasi-experimental comparison group, the evaluator must check for preexisting differences between the groups that could have affected the results (Campbell & Stanley, 1963). For example, the comparison individuals may have had more serious criminal histories than the Drug Court participants to begin with. This, in turn, might have put them at greater risk for criminal recidivism. If so, then superior outcomes for the Drug Court participants might not have been due to the effects of the Drug Court, but rather to the fact that it treated a less severe population. A skilled evaluator can use a number of statistical procedures to adjust for such differences and potentially obtain scientifically defensible results.

Matched Comparison Group—Evaluators do not always have a quasi-experimental comparison group at their disposal. Under such circumstances, they may be required to construct a comparison group out of a large and heterogeneous pool of offenders. For example, an evaluator might need to select comparison subjects from a statewide probation database. Many of those probationers would not have been eligible for Drug Court, or are dissimilar to Drug Court participants on characteristics that are likely to have influenced their outcomes. For example, some of the probationers might not have had serious drug problems, or might have been charged with offenses that would have excluded them from participation in Drug Court. The

evaluator must, therefore, select a subset of individuals from the entire probation pool that are similar to the Drug Court participants on characteristics that are known to affect outcomes. For example, the evaluator might pair each Drug Court participant with a probationer who has the same or similar criminal history, demographic characteristics, and substance use diagnosis (Heck, 2006; Marlowe, 2010, in press). Because the evaluator will choose only those probationers who are similar to the Drug Court participants on multiple characteristics, it is necessary to start out with a large sample of potential candidates from which to select comparable individuals.

The success of any matching strategy will depend largely on whether the evaluator has adequate information about the comparison candidates to make valid matches (Campbell & Stanley, 1963). If data are not available on such important variables as the probationers' criminal histories or substance abuse problems, evaluators and Drug Courts will not be able to place confidence in the validity of the matches. Simply matching the groups on variables that are easy to measure and readily available, such as gender or race, is not sufficient because the groups might differ on other important dimensions that were not taken into account.

Propensity Score Analysis—An evaluator may also use an advanced statistical procedure called a propensity score analysis to mathematically adjust for differences between the Drug Court and comparison groups. This procedure calculates the statistical probability that an individual with a given set of characteristics would be in the Drug Court group as opposed to the comparison group—in other words, the relative similarity of that individual to one group as opposed to the other (Dehejia & Wahba, 2002). The analysis then mathematically adjusts for this relative similarity when comparing outcomes. Advanced statistical expertise is required to implement and interpret this complicated procedure.

As with any statistical adjustment, the success of a propensity score analysis will depend on whether the evaluator has adequate information about the comparison subjects to make valid adjustments. If data are not available on such important variables as the comparison subjects' criminal histories or substance abuse problems, evaluators and Drug Courts will not be able to place confidence in the adjustments (Peikes et al., 2008). Again, merely adjusting the scores based on easily measured variables, such as gender or race, is not sufficient because the groups might differ on other important dimensions that were never taken into account.

Invalid Comparison Groups—Several comparison groups have been used in Drug Court evaluations that quite likely produced seriously biased results. Comparing outcomes from a Drug Court to those of individuals who refused to enter the Drug Court, were denied access to the Drug Court because of their clinical or criminal histories, dropped out of the Drug Court, or were terminated prematurely from the Drug Court is rarely, if ever, justified (Heck, 2006; Heck & Thanner, 2006; Marlowe, 2010, in press; Peters, 1996). The probability is unacceptably high that such persons had poorer prognoses or more severe problems to begin with. For example, they very likely had more serious criminal or substance abuse histories, lower motivation for change, or lesser social supports. Given the high likelihood that these individuals were seriously disadvantaged from the outset, statistical adjustments cannot be relied upon to overcome the differences (Campbell & Stanley, 1963).

J. Time at Risk

For an evaluation to be valid, Drug Court and comparison participants must have the same time at risk, meaning the same opportunity to engage in substance abuse, crime, and other behaviors of interest to the evaluation. If, for example, an evaluator measured criminal recidivism over a period of twelve months for Drug Court participants, but over a period of twenty-four months for the comparison group, this would give an unfair advantage to the Drug Court participants. The comparison group participants would have twelve additional months in which to commit new crimes or other infractions.

Ensuring an equivalent time at risk requires the evaluator to begin the analyses from a comparable start date for both groups. As was mentioned earlier, Drug Court evaluations typically use the date of entry into Drug Court or the date of the arrest or technical violation that made the individual eligible for Drug Court as the start date for analyses. If the comparison group is comprised of probationers, comparable start dates might be the date the individual was placed on probation or the date of the arrest that led to a probation sentence.

If the time at risk differs significantly between groups, the evaluator might be able to compensate for this problem by adjusting statistically for time at risk in outcome comparisons. For example, the evaluator might enter time at risk as a covariate in the statistical analyses. A *covariate* is a variable that is entered first into a statistical model. The independent effect of the variable of interest (in this case, being treated in a Drug Court) is then examined after first taking the effect of the covariate into account. This procedure would indicate whether Drug Court participants had better outcomes after first taking into account the influence of their shorter time at risk. The use of covariates is not always successful, however, and the best course of action is to ensure the groups have equivalent follow-up windows.

A related issue is referred to as *time at liberty*. Time at liberty and time at risk are similar in that both affect a participant's opportunity to reoffend or engage in other behaviors of interest to the evaluation. The difference is that time at liberty relates to whether restrictive conditions were placed on the participant. The most obvious restrictive conditions involve physical barriers to freedom, such as incarceration or placement in a residential treatment facility. These physical barriers severely restrict a participant's ability to use drugs, commit new offenses, obtain a job, or engage in other behaviors of interest to evaluators.

A potential error in Drug Court evaluations is to neglect time at liberty when performing outcome comparisons. In some jurisdictions, for example, individuals who do not enter Drug Court may be more likely to receive a jail sentence. If they are jailed for a portion of the follow-up period, they might have fewer opportunities to reoffend or use drugs than Drug Court participants who are treated in the community. The evaluator might conclude, erroneously, that Drug Court caused participants to reoffend or use drugs more often, when in fact they simply had more time at liberty to do so. Under such circumstances, the evaluator would need to adjust statistically for participants' time at liberty in the outcome analyses. For example, the evaluator might need to enter time at liberty as a covariate in the statistical models. This would indicate whether Drug Court participants had better outcomes after first taking into account their longer time at liberty. As was noted earlier, such adjustments are not always successful and Drug Courts will require expert consultation to ensure the analyses are carried out appropriately.

Note that evaluators are not always advised to adjust for time at liberty. In cost analyses, for example, the time participants spend in jail or a residential treatment facility is an important outcome in its own right and should be valued accordingly from a fiscal standpoint. Deciding whether to adjust for time at liberty, like many evaluation-related decisions, requires scientific expertise and careful consideration of the aims of the study. For such analyses, Drug Courts are strongly advised to obtain expert statistical and scientific consultation.

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APPENDIX C

COMPLEMENTARY NEEDS ASSESSMENTS

This list provides examples of instruments used to assess complementary needs among substance-involved individuals in the criminal justice system. Additional information about needs assessment instruments may be obtained from the following Web sites:

Alcohol and Drug Abuse Institute Library at the University of Washington

<http://lib.adai.washington.edu/instruments/>

The National GAINS Center

<http://gainscenter.samhsa.gov/pdfs/disorders/ScreeningAndAssessment.pdf>

MULTIDIMENSIONAL CLINICAL NEEDS ASSESSMENTS

Addiction Severity Index (ASI)

http://www.tresearch.org/wp-content/uploads/2012/09/ASI_5th_Ed.pdf

Global Appraisal of Individual Needs (GAIN)

<http://www.gaincc.org/products-services/instruments-reports/>

MULTIDIMENSIONAL CRIMINOGENIC NEEDS ASSESSMENTS

Correctional Offender Management Profiling for Alternative Sanctions (COMPAS)

<http://www.northpointeinc.com/products/northpointe-software-suite>

Inventory of Offender Risk, Needs, and Strengths (IORNIS)

<http://www4.parinc.com/Products/Product.aspx?ProductID=IORNIS>

Offender Profile Index (OPI)

<https://www.ncjrs.gov/pdffiles1/Digitization/148829NCJRS.pdf>

Offender Screening Tool (OST)

<http://www.azcourts.gov/apsd/EvidenceBasedPractice/RiskNeedsAssessment/OffenderScreeningTool%28OST%29.aspx>

Ohio Risk Assessment System (ORAS)

http://www.ocjs.ohio.gov/ORAS_FinalReport.pdf

Level of Service/Case Management Inventory (LS/CMI)

[https://ecom.mhs.com/\(S\(0aqkan55ovozwq55w2ox445\)\)/saf_om.aspx?id=Training](https://ecom.mhs.com/(S(0aqkan55ovozwq55w2ox445))/saf_om.aspx?id=Training)

Static Risk and Offender Needs Guide (STRONG)

<https://www.assessments.com/purchase/detail.asp?SKU=5205>

MENTAL HEALTH SCREENS

Beck Depression Inventory-II (BDI-II)

<http://www.pearsonclinical.com/psychology/products/100000159/beck-depression-inventoryii-bdi-ii.html?Pid=015-8018-370&Mode=summary>

Beck Anxiety Inventory (BAI)

<http://www.pearsonclinical.com/psychology/products/100000251/beck-anxiety-inventory-bai.html?Pid=015-8018-400&Mode=summary>

Brief Jail Mental Health Screen (BJMHS)

<http://gainscenter.samhsa.gov/pdfs/disorders/bjmhsform.pdf>

CJ-DATS Co-Occurring Disorder Screening Instrument (CJ-CODSI)

<https://www.drugabuse.gov/sites/default/files/files/CJ-CODSI.pdf>

Generalized Anxiety Disorder 7-Item Scale (GAD-7)

<http://www.integration.samhsa.gov/clinical-practice/GAD708.19.08Cartwright.pdf>

Global Appraisal of Individual Needs-Short Screener (GAIN-SS)

<http://www.gaincc.org/products-services/instruments-reports/>

Mental Health Screening Form-III (MHSF-III)

https://www.idph.state.ia.us/bh/common/pdf/substance_abuse/integrated_services/jackson_mentalhealth_screeningtool.pdf

Modified Mini-Screen (MMS)

<http://www.nyc.gov/html/doh/downloads/pdf/qi/qi-mms-scoringsht.pdf>

Mood Disorder Questionnaire (MDQ)—Bipolar Disorder

<http://www.integration.samhsa.gov/images/res/MDQ.pdf>

Symptom Checklist-90-Revised (SCL-90-R)

<http://www.pearsonclinical.com/psychology/products/100000645/symptom-checklist-90-revised-scl90r.html>

TRAUMA AND PTSD SCALES

Acute Stress Disorder Structured Interview (ASDI)

[http://www.istss.org/assessing-trauma/acute-stress-disorder-structured-interview-\(asdi\).aspx](http://www.istss.org/assessing-trauma/acute-stress-disorder-structured-interview-(asdi).aspx)

Clinician-Administered PTSD Scale for DSM-5 (CAPS-5)

<http://www.ptsd.va.gov/professional/assessment/adult-int/caps.asp>

Life Events Checklist

<http://www.integration.samhsa.gov/clinical-practice/life-event-checklist-lec.pdf>

Posttraumatic Symptom Scale Interview (PSS-I)

<http://www.ptsd.va.gov/professional/assessment/adult-int/pss-i.asp>

PTSD Checklist (PCL)

https://www.facs.org/~media/files/quality%20programs/trauma/vrc%20resources/10_ptsd_checklist_and_scoring.ashx

PTSD Checklist—Civilian Version

http://www.integration.samhsa.gov/clinical-practice/Abbreviated_PCL.pdf

Trauma History Screen

<http://www.ptsd.va.gov/professional/assessment/te-measures/ths.asp>

HEALTH-RISK BEHAVIOR SCALES

HIV Risk Assessment

<http://hivaidsresource.org/hiv-testing/hiv-risk-assessment/>

Texas Christian University (TCU) HIV/AIDS Risk Assessment

<http://ibr.tcu.edu/wp-content/uploads/2014/07/HIV-AIDS-intake-ara.pdf>

University of Pennsylvania Risk Assessment Battery (RAB)

http://www.med.upenn.edu/hiv/rab_download.html

Wisconsin AIDS/HIV Program: Client Assessment Survey

<https://wi-ew.lutherconsulting.com/Wisconsin/commonFiles/downloads/BehavioralRiskSurvey.pdf>

CRIMINAL THINKING SCALES

Criminal Sentiments Scale

https://www.dpscs.state.md.us/publicservs/procurement/QA_5_ATTACHMENT_2_CRIMINAL_SENTIMENT_SCALE.pdf

Texas Christian University Criminal Thinking Scales (TCU-CTS)

<http://ibr.tcu.edu/forms/tcu-criminal-thinking-scales/>

Psychological Inventory of Criminal Thinking Styles (PICTS)

<http://asm.sagepub.com/content/9/3/278.short>

APPENDIX D

EVIDENCE-BASED COMPLEMENTARY TREATMENT AND SOCIAL SERVICES

The following Web sites provide information about evidence-based treatments and social services to address the complementary needs of individuals with substance abuse problems in the criminal justice system.

CLINICAL CASE MANAGEMENT

Case Management Society of America

<http://www.cmsa.org/Home/CMSA/WhoWeAre/tabid/22/Default.aspx>

Commission for Case Management Certification

<http://ccmcertification.org/>

National Treatment Accountability for Safer Communities

<http://nationaltasc.org/resources/>

Treatment Accountability for Safer Communities Crime and Justice Institute, Illinois

<http://www2.tasc.org/>

EVIDENCE-BASED PREVENTION EDUCATION

Bureau of Justice Assistance Naloxone Overdose Reversal Toolkit

<https://www.bjatraining.org/tools/naloxone/Naloxone%2BBackground>

Centers for Disease Control and Prevention (CDCP), HIV/AIDS Prevention Programs

<http://www.cdc.gov/hiv/prevention/programs/index.html>

SAMHSA

Opioid Overdose Prevention Toolkit

<http://store.samhsa.gov/product/Opioid-Overdose-Prevention-Toolkit-Updated-2014/SMA14-4742>

EVIDENCE-BASED TREATMENT AND SOCIAL SERVICES

The Campbell Collaboration Library of Systematic Reviews

<http://www.campbellcollaboration.org/lib/>

The Cochrane Collaboration

<http://www.thecochranelibrary.com/view/0/index.html>

CrimeSolutions.gov

National Institute of Justice (NIJ)

<http://www.crimesolutions.gov/Programs.aspx>

International Society for Traumatic Stress Studies

<https://www.istss.org/>

National Registry of Evidence-Based Programs and Practices (NREPP)

Substance Abuse and Mental Health Services Administration (SAMHSA)

<http://www.nrepp.samhsa.gov/>

APPENDIX E

MANAGEMENT INFORMATION SYSTEMS FOR DRUG COURT EVALUATIONS

This list provides examples of management information systems (MISs) developed to assist in evaluating Drug Courts or other problem-solving courts. Information about additional MISs may be obtained by contacting NDCI faculty or other organizations that perform Drug Court program evaluations.

Buffalo, NY, Drug Court Case Management System (contact the NDCI for more information)

<http://www.ndci.org/contact>

**Advanced Computer Technologies
Drug Court Case Management (DCCM) System**

<http://www.actinnovations.com/solutions/cms.aspx>

Treatment Research Institute Court Evaluation Program (TRI-CEP)

<http://www.tresearch.org/tools/for-courts/tri-cep/demo/>

**Criminal Justice—Drug Abuse Treatment Studies (CJ-DATS)
eCourt System**

<http://www.gmuace.org/documents/prod-pub/cjdats/cjdats-summary-ecourt.pdf>

Social Solutions

Adult Drug Court Performance Model, Efforts to Outcomes (ETO) Software

<http://www.socialsolutions.com/adc/>

Strength Based Digital Connections, LLC

The Virtual File Case Management System for Tribal Courts

www.thevirtualfile.com

